

# Two Years after Pandemic Influenza A/2009/H1N1: What Have We Learned?

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## INTRODUCTION

Previous pandemic influenza viruses involved an antigenic shift to a different subtype. However, the antigenic shift demonstrated by the pandemic influenza A H1N1 2009 virus (A/2009/H1N1) was an antigenic change from a human H1N1 subtype to a swine H1N1 subtype. Pig surveillance programs have not been able to detect the immediate precursor of this virus in pigs in South China and other parts of the world (51). Although South China was predicted to be the starting point of influenza pandemics, the first human case and the initial epidemic were detected in North America and Mexico, respectively (140, 184). Unlike the poor predictability of pig surveillance, human seroepidemiology correctly predicted an impending pandemic due to the lack of immunity in the general population with relative protection in the elderly (387). However, seroepidemiology and laboratory studies were unable to make an accurate assessment of the disease severity

in order to recommend a commensurate pandemic alert level. Regarding the risk factors for severe disease, obesity was an important predisposing factor, in addition to extremes of age, pregnancy, and underlying medical illness (364, 458, 522). In patients with severe disease, viral clearance was delayed, with a persistent elevation of proinflammatory cytokines and associated multiorgan damage despite antiviral therapy (510). Additionally, a lower serum IgG2 level appeared to be associated with disease severity, especially in pregnant patients (80, 207). Severe disease and lung pathology were associated with immune complex deposition. In terms of laboratory diagnosis, a comparative laboratory test eval-

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doi:10.1128/CMR.05012-11

uation showed that a rapid diagnosis was best achieved by reverse transcription-PCR (RT-PCR), which was markedly superior to antigen detection by enzyme immunoassays (EIAs) (296). None of the viral genomic signatures, such as PB2-K627, full-length PB1-F2, and the PDZ motif of NS1, which were previously speculated to be useful in predicting the virulence of the influenza A virus were present in this pandemic virus (293). Only the hemagglutinin D222G (H1 numbering) mutation with a predilection for  $\alpha$ -2,3-linked sialic acid was associated with clinically severe disease and the involvement of the lower respiratory tract (96). In addition, the similar crystal structures, antigenic makeups, and patterns of glycosylation for the hemagglutinin of this virus and that of the 1918 virus explained the relative protection of the elderly through the induction of cross-reactive humoral and cellular immune responses against the surface and internal proteins, respectively. As for treatment options, the intrinsic resistance to adamantanes leaves the early initiation of neuraminidase inhibitors as the only option available in most countries. Moreover, further mutations may affect the usefulness of these antivirals. In severe cases, intravenous peramivir or zanamivir, convalescent-phase plasma, and hyperimmune intravenous immunoglobulin can be considered in clinical trial settings. Despite the technological advances in using cell-based inactivated whole-virus vaccines and improved adjuvants, vaccine production failed to prevent the first peak in tropical areas and the Southern Hemisphere. The bottlenecks for the rapid mass production of vaccines must be overcome before the next pandemic. Social distancing methods, such as canceling entertainment and sporting events, closing stores, office buildings, and public transportation systems, border screening, the isolation and quarantine of febrile patients and contacts, school closures, and hospital infection control measures may achieve only a few more weeks of preparedness by slowing down the introduction and spread of the pandemic virus if instituted early enough. In this article, we review the biology of the virus in relation to the clinical manifestations, pathogenesis, laboratory diagnosis, host susceptibility, immune response, and options for treatment, immunization, public health, and infection control. Because there have been a large number of publications on this topic, we can select only those publications related to the understanding and practice of clinical microbiology and infectious diseases.

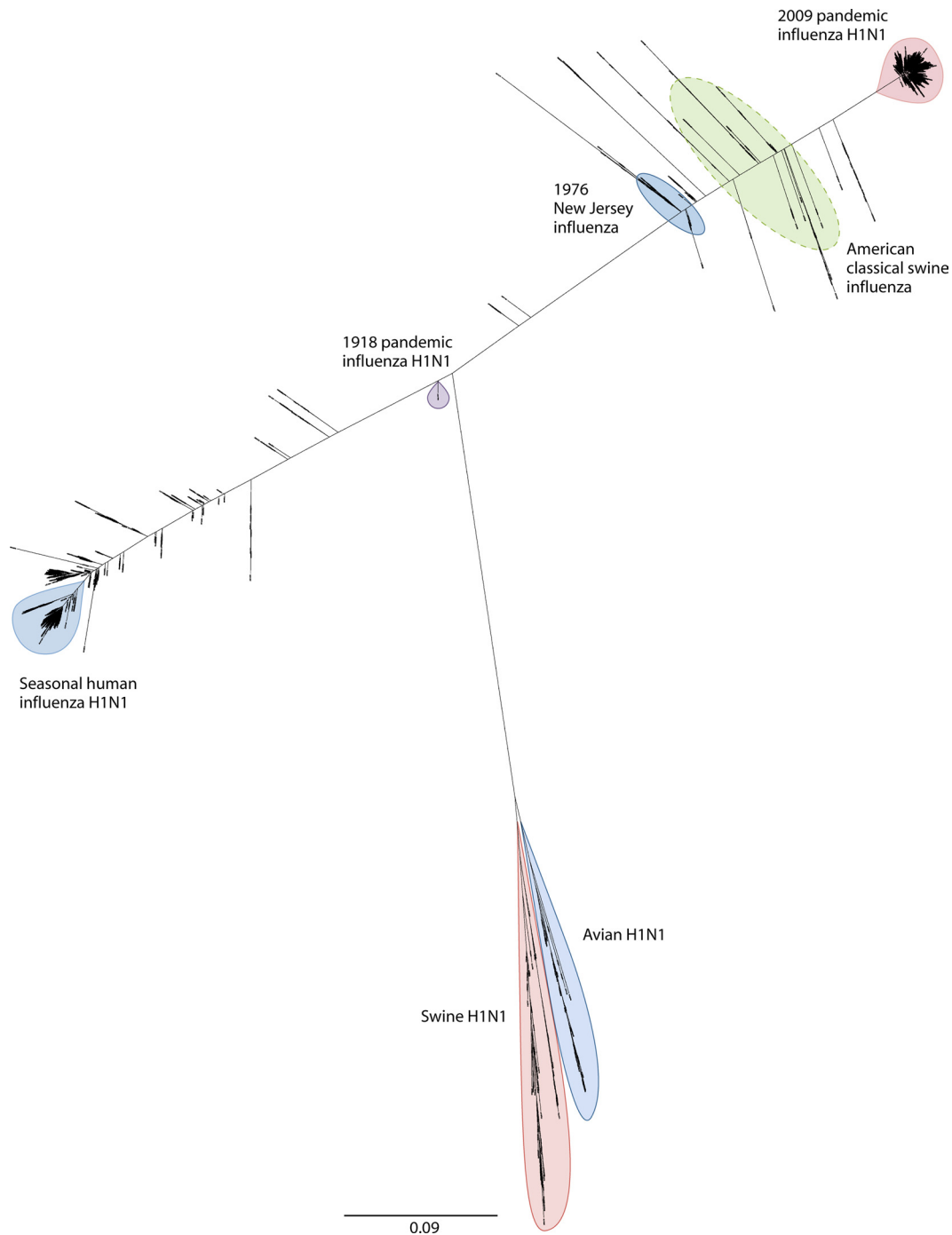
### TAXONOMY, NOMENCLATURE, AND GENERAL VIROLOGY

The influenza virus is a negative-sense, single-stranded RNA virus with a lipid-containing envelope and an eight-segment genome encoding 11 or 12 proteins (545). The pleomorphic viral particles observed in cell culture or clinical specimens may vary from spherical to filamentous in shape, with a diameter of 120 nm. The A/2009/H1N1 virus produces filamentous viral particles when grown in cell lines examined by electron microscopy (274). This virus belongs to the family *Orthomyxoviridae*, which includes the genera of influenza virus, Thogoto virus, and infectious salmon anemia virus (155, 424). The genus of influenza virus is divided into types A, B, and C as defined by the antigenicity of the nucleocapsid and matrix proteins in the viral core (424). These proteins are the antigens used in the complement fixation test for type-specific antibodies (250). Among the three genera, the influenza A virus is associated with more severe disease and pandemics in humans (234, 569). The influenza A virus is further subtyped by two surface proteins, hemagglutinin (H), which attaches the virion to the host cell for cell entry, and neuraminidase (N), which

facilitates the spread of the progeny virus by cleaving the host sialic acid receptors attached to the progeny virus (137). There are 16 H subtypes and 9 N subtypes, which make up all of the subtypes of the influenza A virus by various combinations of H and N (180). The infidelity of RNA polymerase (one mutation per genome per replication) and the selective pressure of host immunity lead to the accumulation of mutations and a change in the surface antigenicity of these surface proteins, which are the targets of neutralizing and hemagglutination-inhibiting antibodies (2). This antigenic change is called antigenic drift (424). Thus, every year in February, the World Health Organization (WHO) selects the strains of virus that are to be used for the annual influenza vaccination in humans according to this antigenic drift. The nomenclature of the viral strain is written in the following order: type/place of isolation/strain number/year of isolation(subtype). Using the year 2005 as an example, the WHO recommended changing the virus strain from influenza A/Fujian/411/2003(H3N2) to A/California/7/2004(H3N2) for vaccination purposes. These changes are necessary because the hemagglutination inhibition antibody titers in the general population that would be induced by the older vaccine strain would not be sufficient to provide good protection against a newly drifted or, in the case of a pandemic, newly shifted strain (429). All of the combinations of the 16 H and 9 N viral subtypes are found in water fowl, whereas the H1 to H3 and N1 and N2 viral subtypes are commonly found in humans infected with influenza (4). Because of its segmented genome, the shuffling of gene segments can occur if two different subtypes of the influenza A virus infect the same cell. This genetic reassortment of the gene segments from different animals can lead to a major surface antigenic change, resulting in a new reassorted virus to which the global population has no effective neutralizing antibodies. A pandemic may ensue if this new virus has preserved replicative efficiency in human cells and transmissibility between humans.

A phylogenetic analysis of the A/2009/H1N1 virus suggested that the ancestors of the 8 gene segments are different and can be traced to avian, human, and swine origins (481). In previous pandemic viruses, the H subtypes shifted from H1 to H2 in 1957 and H2 to H3 in 1968, whereas the 2009 pandemic virus had North American swine H1 and Eurasian swine N1, replacing the circulating human seasonal H1N1 virus subtype (472). Using all of the available H1N1 virus sequences in GenBank, phylogenetic analysis suggested that the A/2009/H1N1 virus strains, and the 1976 New Jersey swine strains, and the 1918 pandemic H1N1 strains are distinct and diverge from the human seasonal H1N1 strains circulating since 1918 (Fig. 1). This form of genetic shift is different from those of the 1957 pandemic H2N2 virus, which evolved from the circulating H1N1 virus by acquiring the H2, N2, and PB1 genes from avian species (301), and the 1968 pandemic H3N2 virus, which is likely to have evolved from the circulating H2N2 virus by acquiring the H3 and PB1 genes from avian species (173, 301). Thus, animal surveillance programs for influenza virus will continue to play an important role in the detection of the immediate precursor virus in animals before it becomes a pandemic virus in humans. However, the vast geographic distribution of pig and poultry farming, especially in developing countries, presents an enormous challenge for surveillance programs to detect such a precursor virus before a pandemic begins.

The 8 gene segments of the influenza A virus genome, encoding 11 to 12 viral proteins with various functions in the life cycle, often

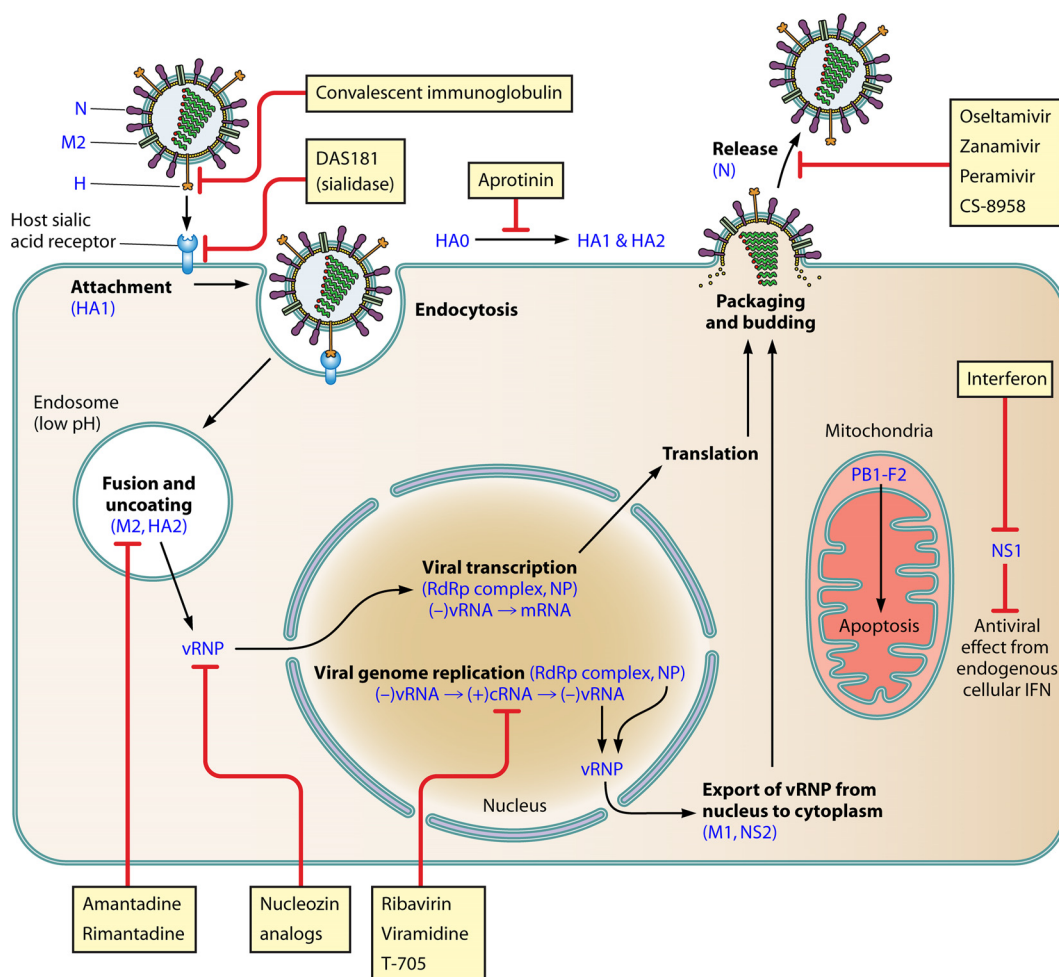


**FIG 1** Maximum-likelihood phylogenetic tree of 3,764 full-length hemagglutinin nucleotide sequences of A/2009/H1N1 strains from the NCBI Influenza Virus Resource. Phylogenetic reconstruction was performed using RAXML version 7.2.6. Positions of 1918 pandemic, 1976 New Jersey swine influenza, seasonal, and 2009 pandemic human influenza A virus strains are highlighted.

serve as antiviral and diagnostic targets (Fig. 2). Three gene segments encode the polymerase proteins PB1, PB2, and PA and two smaller proteins, PB1-F2 and N40. The polymerase proteins are antiviral targets of ribavirin, viremide, and T-705 (141, 312). PB2 is necessary for cap binding and endonuclease activity but can also inhibit the induction of type I interferon (276). PA functions as an RNA polymerase subunit and in proteolysis (441). PB1-F2 is a

mitochondrial toxin that causes cellular apoptosis and inhibits the induction of type I interferon (524). Additionally, a newly identified protein of unknown function, N40, is expressed from the PB1 fragment (545).

Nucleoprotein (NP) can be an antiviral target of nucleozin and its analogs (295). NP is the structural component of ribonucleoprotein (RNP). H is the target for neutralizing antibodies and has



#### Diagnostic test (include antiviral susceptibility testing)

- H – Hemagglutination inhibition and viral neutralizing antibody assay for strain- or subtype-specific antibody; D222G (marker of higher virulence); precursor polypeptide HA0 is proteolytically cleaved into HA1 and HA2 for cell entry.
  - N – Oseltamivir resistance (H275Y); zanamivir resistance (S247N, I223R, I223K)
  - M2 – Amantadine resistance (S31N)
  - M1, NP – RT-PCR and type specific antibody testing by complement fixation test for type specific antibody
- Inhibitory action

FIG 2 Potential diagnostic and antiviral targets in the viral life cycle of influenza virus.

a precursor form called HA0 that undergoes proteolytic cleavage into two components, HA1 and HA2. HA1 functions in the attachment of the virus onto the sialic acid receptor, and HA2 is responsible for membrane fusion. H has 4 major antigenic domains, with frequent mutations in 5 hypervariable regions at HA1. N is the antiviral target of oseltamivir, zanamivir, and peramivir (243). N serves as a sialidase that is required for the release of progeny virions (425). N binds plasminogen for the activation of H by proteolytic cleavage (208). The matrix protein M1 is a structural protein for the nuclear export of viral RNA (vRNA) and viral budding, whereas M2 functions as an ion channel for the acidification of the viral core during the uncoating of virus in endosomes. M2 is the antiviral target of the adamantanes, which include amantadine and rimantadine. In addition, M2 induces heterotypic antibodies associated with protection. The nonstructural protein NS1 is an interferon antagonist that may also stimulate proinflammatory cytokines in infected cells, whereas NS2 is a nuclear export factor.

In addition to the spikes in the H and N proteins that cover the surface of the virion (402), the M2 protein is embedded in the lipid envelope with its ectodomain exposed on the surface (444). Below the lipid envelope is a layer of the M1 protein that surrounds the RNP core (573). The core consists of 8 RNA segments that are associated with one or more copies of the viral polymerase complex and covered by NP molecules (16). Recent studies suggested that the genetically highly conserved M2 ectodomain and the stalk region of HA2 could be the antigenic targets of neutralizing antibodies for the production of a universal influenza A vaccine (37, 483). Glycosylation has been associated with the masking of epitopes on H, and the A/2009/H1N1 virus has significantly reduced glycosylation compared to the previous seasonal influenza H1N1 virus (563). This finding may partly explain why a single dose of the A/2009/H1N1 vaccine was sufficient enough to induce immunity in immunologically naïve individuals.

The pandemic influenza virus behaves like other influenza viruses, which are infectious for more than 24 h under low (<25%)



and high (>80%) relative humidity but less stable at intermediate humidity (50%) and higher temperatures (366, 490). Thus, the virus should be most stable in cold winters. Unexpectedly, the A/2009/H1N1 virus emerged in the spring in North America and spread during the summer in tropical areas and the Southern Hemisphere. This virus produced a larger second wave during the early autumn in the Northern Hemisphere. Thus, the pressure of seasonality on transmission was overwhelmed by the lack of herd immunological control. The virus can be readily inactivated by detergents, disinfectants, ionizing radiation, or temperatures greater than 50°C (287).

## VIRAL LIFE CYCLE

The virus initiates its life cycle when H attaches to the receptors on a cell surface that contains sialic acid, such as glycolipids or membrane glycoproteins. Human influenza viruses, including the A/2009/H1N1 virus, preferentially bind to sialic acid with an  $\alpha$ -2,6-linkage to galactose, which contains oligosaccharides that are abundantly present in the upper respiratory tract (473). In contrast to the other human seasonal influenza A/H1N1 virus, the A/2009/H1N1 virus also binds to the  $\alpha$ -2,3-linked sialic acid receptors that are present in the lower respiratory tract (113). The internalization of the virus into endosomes occurs by endocytosis, which is mediated by epsin-1, a cargo-specific adaptor for virus entry through the clathrin-mediated pathway or by the clathrin-independent pathway (91). The precursor polypeptide HA0 must be activated by proteolytic cleavage into HA1 and HA2 by trypsin or a trypsin-like endogenous host cell protease before it can undergo an acid pH-triggered conformational change into a fusogenic form (97). Then, the viral and endosomal membranes can be fused. The ion channel M2 in the viral membrane is then activated by endosomal acid pH, resulting in an influx of protons into the virion and the dismantling of M1 from the RNP core (465). The dismantling process will release the RNP into the cytoplasm. PB2 and NP both have a nuclear localization signal (NLS) and can bind to importin  $\alpha_1$  (a cellular nuclear import factor), the most abundant importin in human cells (189). Next, the RNP enters through the nuclear pore into the nucleus, where the transcription and replication of the viral genome occur (254). This nuclear localization of influenza virus NP results in the egg yolk appearance of an infected cell that is stained by an immunofluorescent antibody against the NP in clinical specimens, such as nasopharyngeal aspirate. Moreover, blocking the trafficking of the viral NP from the cytoplasm into the nucleus by nucleozin analogs was shown to be a viable antiviral mechanism (295). The nuclear viral RNP then becomes the template for the production of mRNA that snatches its 5' cap of 9 to 15 nucleotides from host mRNA (320, 460). In addition, this viral RNP is the template for the full-length complementary copy that becomes a template for the amplification of viral RNA genomes. With an NLS, viral PB1, PB2, PA, and NP are transported into the nucleus, where they associate to form RNP (290, 397, 406, 413). Both vRNA and cRNA complexed with RNP have a panhandle/fork/corkscrew structure (178). The export of viral RNP back to the cytoplasm depends on M1, which binds to the RNP and interacts with NS2, which has a nuclear export signal (471). Under the cytoplasmic membrane, the RNP-M complex assembles under patches of H and N to form virions (499). The packaging of vRNA in viral particles depends on the 3' and 5' noncoding regions with *cis*-acting signals (268). Additionally, the coding sequences of the viral RNA contribute to the efficient pack-

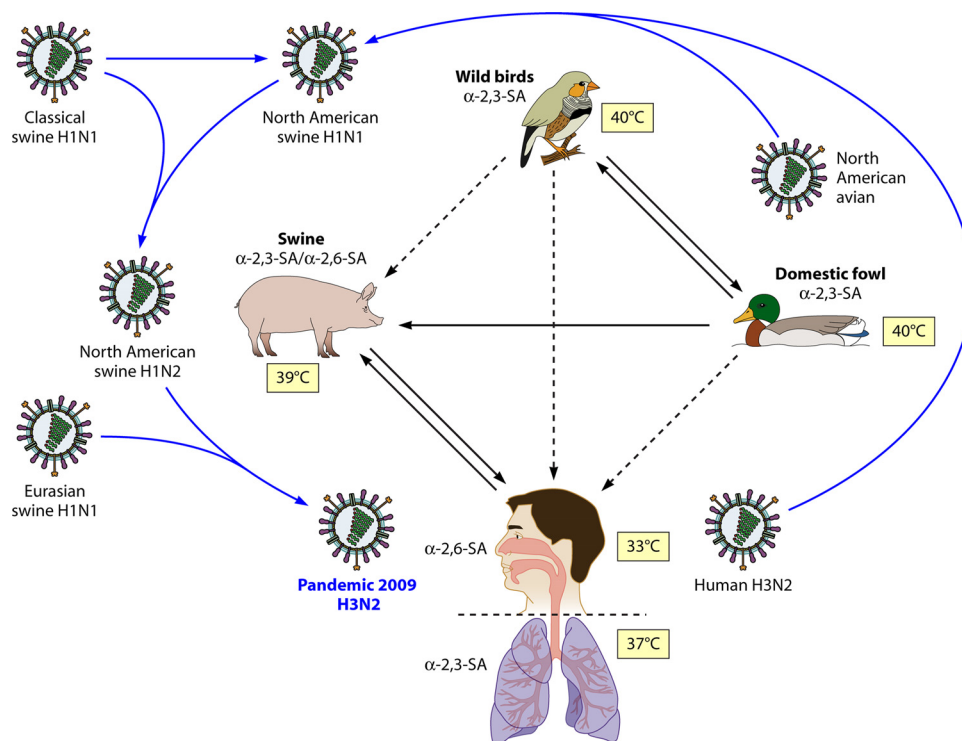
aging, but the exact mechanism is unknown (401). The budding of infectious progeny virions is followed by their release from their aggregation by their viral H bound to sialic acid, which is mediated by the sialidase activity of N. Because the mRNAs of H, M, NP, and NS are more abundantly expressed than the PA, PB1, PB2, and N genes, RT-PCR diagnostic tests are often targeted against the M gene, which is abundant and genetically conserved, and the H gene, which is abundant and subtype specific (153, 331, 447). The most abundant protein that is expressed in infected cells and constitutes the structure of a free virion is the NP (448). Thus, most EIAs for the influenza virus antigen are targeted toward the NP, which is type specific, but these tests cannot differentiate between subtypes.

## CELL ENTRY AND VIRULENCE FACTORS

Virus-containing droplets may settle on nasopharyngeal, tracheobronchial, conjunctival, or other respiratory mucosal epithelia. The virus moves randomly through the mucus layer of the mucosa with the aid of neuraminidase to find the appropriate cell receptor for binding by viral H. The  $\alpha$ -2,6-linked sialic acid receptors are present in both the upper and lower respiratory tracts of humans, the latter of which are lined by tracheobronchial pseudocolumnar epithelium and type 1 pneumocytes. The  $\alpha$ -2,3-linked sialic acid receptors are present in the distal bronchiole, type 2 pneumocytes, and alveolar macrophages of humans (473). The H of the A/2009/H1N1 virus adheres to the  $\alpha$ -2,6-linked sialic acid receptors and to the  $\alpha$ -2,3-linked sialic acid receptors, which are more highly expressed in the human lower respiratory tree. The latter binding affinity is similar to that of the H subtypes of the avian influenza A virus, such as the H5N1, H9N2, or H7N7 subtype, which also preferentially attach to the  $\alpha$ -2,3-linked sialic acid receptors that are present on the respiratory and alimentary epithelia of birds, human conjunctiva, and the ciliated portion of human respiratory pseudostratified columnar epithelium; therefore, this binding affinity allows for the sporadic jumping of avian influenza viruses into humans (Fig. 3). Moreover, this binding affinity allows the A/2009/H1N1 virus to infect and replicate to high titers in the lungs of ferrets, mice, and monkeys, with a greater proinflammatory response than for seasonal influenza virus. Nevertheless, the damage caused by the A/2009/H1N1 virus is less than that of the H5N1 virus (274, 369, 400, 520). This finding may partially explain the predilection of this new virus to cause pneumonia in healthy individuals (113). However, this *in vitro* glycan binding phenomenon has not been replicated in other studies (41, 99, 354, 369, 564, 567).

In addition to its pathogenetic significance, the differential binding affinity of H for different receptors affects the evolution of influenza virus. The respiratory epithelium of pigs contains both  $\alpha$ -2,6- and  $\alpha$ -2,3-linked sialic acid receptors; therefore, pigs can be infected by human, swine, and avian influenza viruses. Because pigs live in close proximity to humans and poultry in residential backyards in Southeast Asia, they can theoretically serve as "mixing vessels" for gene reassortment between avian, swine, and human influenza viruses, which may create future pandemics.

The conserved amino acid residues within the influenza virus H receptor binding site that are implicated in receptor specificity include amino acid positions 98, 136, 153, 183, 190, 194, 222, 225, 226, 227, and 228. Remarkably, the only difference between the 1918 H1 of the A/New York/1918/H1N1 virus and an avian H consensus receptor binding signature is a single E190D mutation



**FIG 3** Emergence of A/2009/H1N1 virus from other human and animal influenza viruses. Influenza viruses are usually limited to infecting specific hosts, with tissue tropism and receptor specificity being important restriction factors. A change in tropism sometimes occurs, with pigs being an important “mixing vessel” due to their tracheae containing receptors with both  $\alpha$ -2,3-linked and  $\alpha$ -2,6-linked sialic acid moieties. In the case of A/2009/H1N1 virus, sequential reassortment of genes from human, avian, and swine influenza viruses culminates in a virus with replication competence comparable to those of other human influenza viruses.

(H3 numbering) (476). The H crystal structure of the A/2009/H1N1 virus reveals a close resemblance to the 1918 H1N1 virus (563). The H1 of the A/South Carolina/1918/H1N1 virus has an additional G225D (H3 numbering) substitution, which is sufficient to switch the receptor preference from  $\alpha$ -2,3 to  $\alpha$ -2,6 in cell-based assays (501). The A/2009/H1N1 virus possesses a “human virus”-type amino acid at positions 190 and 225 (222 by H1 numbering), which most likely supports the efficient transmissibility of these viruses in humans. In addition to these two positions, a difference in only two other amino acids, at positions 200 and 227, differentiated the receptor binding specificities of the swine H1 virus and the pandemic H1 virus (149). When the A/2009/H1N1 virus was compared with the classical swine virus, the most discriminative positions were all located in the receptor binding pocket, suggesting that these positions may be important in the human adaptation of the virus (386). Interestingly, some isolates from patients with severe disease, especially in lower respiratory tract specimens, possess an amino acid substitution at position D222G (H1 numbering) (96), and D222G is associated with viremia (516). The D222G mutant has been readily generated in mouse adaptation experiments that demonstrated a significant increase of virulence in mice (580). Additionally, the D222G mutant has been shown to have stronger binding to turkey erythrocytes than wild-type virus (85). Homology modeling and molecular docking showed that the receptor binding pocket of the A/2009/H1N1 virus is smaller than those of other influenza A viruses, which allows for a tighter binding of the virus with the receptor. The amino acid at position 222 of H may affect the positioning of the conserved Q223 residue, hence modulating the

flexibility of the binding pocket and steric hindrance during receptor binding. In addition, the molecular property of residue 222 can directly influence the “lysine fence” via the polarity of the amino acid residue, where the D222G substitution enhances the electrostatic interactions between the receptor and the protein (515). Furthermore, the lysine fence at the H of the A/2009/H1N1 virus, which is a positively charged structure composed of K145, K133, and K222, favors the binding of the virus to both  $\alpha$ -2,3- and  $\alpha$ -2,6-glycans (487).

Viruses that possess PB2-K627 but not viruses that possess PB2-E627 were previously observed to grow efficiently in the upper respiratory tracts of mammals (241), which suggests that PB2-K627 confers efficient replication at 33°C (the temperature of the upper airway in humans), whereas PB2-E627 does not. In contrast, both variants mediate efficient replication at 37°C. Collectively, these findings suggest that PB2-K627 allows for efficient replication in the lower and upper respiratory tracts of mammals, a feature that may facilitate transmission. Notably, the replacement of PB2-K627 with glutamic acid reduced the transmissibility of human influenza viruses in a guinea pig model (489). In addition, the amino acid substitution at position 701 of PB2 has emerged as a determinant of virulence, a role probably related to its facilitation of the binding of PB2 to importin in mammalian cells. The recently emerged A/2009/H1N1 virus possesses “low-pathogenic”-type amino acids at positions 627 and 701, i.e., glutamic acid and aspartic acid, respectively (426). However, these low-virulence pathotypes may be overcome by PB2-R591, which has been shown to confer efficient replication of the virus (566).

Other virulence-associated genomic signatures, such as the

full-length PB1-F2, the cleavage and polyadenylation specificity factor 30-kDa subunit (CPSF30), the E92 and PDZ domains of NS1, and the multibasic H cleavage sequence, which were previously speculated to be useful in predicting the severity of disease, are absent in the pandemic A/2009/H1N1 virus (193). Most virus strains have a truncated version of PB1-F2 that consists of 11 amino acids. Three isolates of the A/2009/H1N1 virus were reported to have PB1-F2, which consisted of 57 amino acids and was associated with an improvement in *in vitro* replication (421). In cell culture and in an animal model, the introduction of various virulence-associated mutations into PB2 generally did not enhance the viral titer or virulence of the A/2009/H1N1 virus (252). The only consistent viral signature associated with clinical severity, higher viral titer, and virulence in animal models is the D222G mutation (Table 1), which is likely due to its high affinity for binding to  $\alpha$ -2,3-linked sialic acid receptors in the lower respiratory tract. However, these mutants may have diminished transmissibility between the upper respiratory tracts from host to host, which may have selected against their prevalence.

A comparison of the influenza viruses from different origins found that the residues E14 and F55 in the M2 proteins were predominantly in the human influenza virus but not in the swine or avian influenza viruses, suggesting that these residues may be important in human-to-human transmission (427). The M1 protein has been shown to be essential in the high transmissibility of the A/2009/H1N1 virus in a guinea pig model (118). Furthermore, a genomic analysis of the A/2009/H1N1 strains that were collected during the early prepandemic period showed that there have been multiple changes in the viral genome (382), and several mutations in NP (V100I), NS1 (I123V), N (V91I and N233D), and H (S206T) have emerged as the predominant type, from fewer than 11% of the strains collected from April 2009 to more than 75% of the strains collected from June to December 2009 (426). Because most of these mutant residues were located in the functional domain, it was proposed that these changes may enhance human adaptation of the virus or its virulence.

## EPIDEMIOLOGY

When the chronology of the 2009 pandemic was examined (Table 2), it was found that there were many similarities with the severe acute respiratory syndrome (SARS) epidemic (105, 106, 109, 437, 438), except that the 2009 pandemic was considered to be mild in the first wave (449). The outbreak started in a developing country, Mexico, but the virus was first isolated in California, where the public health and laboratory infrastructures are well established (140). Although a phylogenetic analysis strongly suggested that swine would be the immediate animal host from which the virus jumped into humans, the immediate virus precursor could not be detected by animal surveillance programs. However, in analogy to the bat SARS coronavirus (332), a more distant precursor virus was found in swine. Fortunately, for the SARS epidemic, the immediate precursor virus of the human SARS coronavirus was found in civets, which were later banned in wet markets to prevent the relapse of the SARS epidemic (216). Paradoxically, the retrograde passage of the A/2009/H1N1 virus back into pigs was well documented with genetic reassortment (530). Though oseltamivir resistance or D222G viral mutation associated with severe disease was reported within 9 months, the proportions were very low (372). In contrast to the SARS coronavirus, which could be controlled by infection control measures and social distancing, the

A/2009/H1N1 virus rapidly infected over 40% of the susceptible population and became endemic by replacing the previously circulating human seasonal H1N1 virus.

## Animal Surveillance

The scientific community anticipates that a future influenza pandemic is likely to originate from animals; however, many scientists were surprised that the 2009 pandemic may have originated from swine (481) and not from avian populations, which had been severely affected by the H5N1 and H9N2 viruses since 1997, with occasional poultry-to-human transmission (25, 52, 103, 104, 214, 215, 439, 440, 576). In the prepandemic period, intensive screening was dedicated to poultry surveillance, especially in East Asia, where the lethal H5N1 first emerged (94, 214, 215, 351, 576) and where antigenically drifted H3N2 strains often emerged and spread throughout the world (462). There were several warning signals for the 2009 pandemic. First, a triple-reassortant swine influenza H1N1 virus that represented a major antigenic shift event, composed of genomes from the H1N1, H3N2, and H1N2 viruses, emerged in the swine population in the late 1990s (417). Second, this reassortant virus repeatedly jumped the species barrier and caused sporadic cases in humans, with 20% of these cases requiring invasive mechanical ventilation (408, 472). Increased surveillance of pigs was proposed immediately before the pandemic (212). However, the vast geographic extent of pig farming, especially in developing areas, makes pig surveillance difficult.

## Mathematical Modeling

Mathematical modeling has been widely used to predict the spread of influenza virus and the effectiveness of containment strategies (342). Although it is useful as a conceptual model, prepandemic modeling assumed that the pandemic began in Southeast Asia (120, 174). During the current pandemic, real-time predictions of the spread of the pandemic have been performed but with only modest accuracy (19, 176). Mathematical modeling can estimate the worst- and best-case scenarios to gauge the magnitude of the pandemic only when a sufficient caseload is available to calculate the basic parameters for the modeling. However, the interval for data accumulation has diminished the predictive value of mathematical modeling and its impact on epidemiological control. There have been few examples of policy changes or epidemiological controls that were based directly on the findings of mathematical modeling.

Epidemiological studies of laboratory-confirmed cases of 2009 pandemic influenza (Table 3) showed that for community outbreaks, the clinical attack rate ranged from 5.4% to 20.6%, the reproductive number ranged from 1.1 to 3.1, the generation time ranged from 0.8 to 3.9 days, and the incubation period ranged from 2 to 5 days (20, 36, 65, 134, 154, 184, 205, 226, 262, 359, 416, 506, 543, 556, 558, 571). These figures did not differ from those in the epidemiological settings of households, schools, hospitals, and tours. No clear differences were observed in the studies conducted across different continents.

## Seroepidemiology

Seroepidemiological studies have demonstrated that children and young adults were seronegative before 2009 and were more commonly infected than the elderly after the first wave (387). Compared with patients with seasonal influenza, those with the A/2009/H1N1 virus were younger and less likely to have underly-

TABLE 1 Viral mutations and characteristics associated with virulence of A/2009/H1N1 influenza virus<sup>a</sup>

| Viral protein | Study design                                                                                                                                                                 | Finding(s)                                                                                                                                                                                                                                                                         | Reference |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| PB2           | Ferret model; introduction of E627K and D701N into A/2009/H1N1                                                                                                               | No enhancement of virulence                                                                                                                                                                                                                                                        | 252       |
|               | Cell culture (A549) and mice; introduction of E627K and D701N into A/2009/H1N1                                                                                               | Both mutants grew to 1-2-log-lower peak titer than parental strain in A549 and had attenuated viral replication in mouse lung; E627K mutant displayed only rare focal inflammation, compared with focal bronchiolitis caused by parental strain                                    | 280       |
|               | Human (293T) and porcine cells; to study the mechanism for human adaptation for A/2009/H1N1 which carries avian type E627 in the PB2 subunit                                 | Serine at position 590 and arginine at position 591 found important in the regulation of polymerase activity in human cells                                                                                                                                                        | 381       |
|               | Mouse and ferret model; to study the effect of varying the amino acids at positions 591, 627, and 701 of PB2 in A/2009/H1N1                                                  | R591Q reduced polymerase activity, while E627K and D701N had only mild effects; in competition studies between wild-type virus and R591Q mutant, relative amounts of wild-type virus increased over time in infected ferrets                                                       | 566       |
|               | Cell culture (MDCK, 293T, DF-1, A549)                                                                                                                                        | 271A responsible for high polymerase activity of the virus                                                                                                                                                                                                                         | 50        |
|               | Mouse model; analysis of the effect of amino acid substitution                                                                                                               | E158G/A enhanced transcription and replication activity                                                                                                                                                                                                                            | 271       |
|               | Mouse and ferret model; to study the importance of position 591 of PB2                                                                                                       | A basic amino acid at position 591 of PB2 can compensate for the lack of lysine at position 627 in pH1N1 for efficient viral replication in mammals                                                                                                                                | 566       |
|               | Mouse model; selection of virulent mouse strain and comparison of the PB2 gene                                                                                               | Position 158 is a pathogenic determinant                                                                                                                                                                                                                                           | 584       |
|               | Mouse model                                                                                                                                                                  | Substitution of 627K in PB2 gene does not confer higher virulence                                                                                                                                                                                                                  | 589       |
| H             | Compare patients with fatal and nonfatal cases (worldwide)                                                                                                                   | Prevalence of D222G in fatal vs nonfatal cases (7.1% vs <1.8%); D222G enhanced binding to $\alpha$ -2,3-linked sialic acid cell receptors                                                                                                                                          | 553       |
|               | Analyze 189 respiratory specimens from influenza surveillance network in Spain                                                                                               | D222G found in 2 fatal cases and 1 patient with severe illness who recovered; D222E found in 61 samples; no mutations found at position 187                                                                                                                                        | 12        |
|               | Analyze 117 clinical specimens in Hong Kong                                                                                                                                  | D222G found in 12.5% of patients with severe disease but not in patients with mild disease; D222G identified mainly in endotracheal samples; D222N found in 2 severe cases and in 1 mild case                                                                                      | 96        |
|               | Analyze 11 specimens from 5 deceased patients and 9 specimens from 5 ICU patients in Canada                                                                                  | D222G found in 3 ICU patients but none in deceased patients; D222G was more frequent among lower respiratory tract specimens than nasopharyngeal specimens (45% vs 14%; $P = 0.18$ )                                                                                               | 156       |
|               | Analyze 61 severe cases and 205 mild cases in Norway                                                                                                                         | D222G found in 11/61 severe cases but not found in any mild cases ( $P < 0.001$ ); D222N was found in 3 severe cases and 1 mild case ( $P = 0.039$ ); no difference in the frequency of D222E between severe and mild cases ( $P = 0.072$ )                                        | 304       |
|               | Analyze 219 severe and 239 nonsevere cases in Hong Kong                                                                                                                      | D222G found in 4.1% of severe cases, but not found in nonsevere cases ( $P = 0.002$ )                                                                                                                                                                                              | 371       |
|               | Analyze 8 fatal cases, 23 severe pneumonia cases, and 13 mild cases in Greece                                                                                                | D222G found in 5/31 (16.1%) severe/fatal cases and 2/13 (15.4%) mild cases; S162R found in 1 patient with severe illness                                                                                                                                                           | 382       |
|               | Analyze 23 fatal cases, 9 seriously ill cases, and 26 community cases in UK                                                                                                  | D222G found in 8.7% of fatal cases but not found in seriously ill or community cases; D222N found in 22% of seriously ill patients but not found in fatal or community cases; no difference in frequency of D222E between severe and mild cases                                    | 388       |
|               | Analyze 130 patients (23 with severe disease) in Italy for D222G and describe the transmission of A/2009/H1N1 with D222G                                                     | Only 1 patient with severe disease infected with D222G virus; this virus was transmitted to the father of the index patient, who was "moderately ill" without hospitalization                                                                                                      | 451       |
|               | Mouse and ferret model                                                                                                                                                       | D222G caused ocular disease in mice but no enhanced virulence in mice or ferrets; D222G attached to a higher proportion of alveolar macrophages and type II pneumocytes                                                                                                            | 119       |
|               | Analyze 63 patients with A/2009/H1N1 infection                                                                                                                               | D222G in 17.4% of severe pneumonia and in 26.7% of ICU patients                                                                                                                                                                                                                    | 89        |
|               | Analyze the H of A/2009/H1N1 from 273 severe cases and 533 nonsevere cases in Spain                                                                                          | D222G present in 5.12% of severe cases; D222E present in 17.21% of severe cases and 9.75% of nonsevere cases; D222N found in 3 severe cases                                                                                                                                        | 337       |
|               | Mouse model; to study the effect of D222G substitution on virulence                                                                                                          | D222G was more lethal in chicken embryo and produced higher viral load; the mutant had a much lower 50% lethal dose than wild-type virus ( $1.5 \times 10^2$ PFU vs $2 \times 10^6$ PFU); mortality of mice due to mutant virus higher than that due to wild type ( $P < 0.0001$ ) | 580       |
|               | Mouse model; to compare the D222G mutant with wild-type virus                                                                                                                | D222G caused more severe disease in both nonpregnant and pregnant mice                                                                                                                                                                                                             | 85        |
|               | Analyze the receptor binding characteristics by carbohydrate microarray analyses (neoglycolipid technology)                                                                  | A/2009/H1N1 (A/California/4/2009 and A/Hamburg/5/2009) bound to both $\alpha$ -2,6- and $\alpha$ -2,3-linked sialyl sequences                                                                                                                                                      | 113       |
|               | Infection of <i>in vitro</i> culture of human airway epithelium (HTBE) with D222G mutant virus; examine receptor binding characteristics by carbohydrate microarray analyses | D222G virus infected a higher proportion of ciliated cells; also bound to a broader range of $\alpha$ -2,3-linked sialyl sequences                                                                                                                                                 | 360       |
|               | Analyze the receptor binding characteristics by use of recombinant H                                                                                                         | A/2009/H1N1 binds only $\alpha$ -2,6-linked and not $\alpha$ -2,3-linked sialyl sequences                                                                                                                                                                                          | 567       |
|               | Analyze the receptor binding characteristics by use of recombinant H                                                                                                         | A/2009/H1N1 binds $\alpha$ -2,6-linked but only minimally binds $\alpha$ -2,3-linked sialyl sequences                                                                                                                                                                              | 369       |
|               | Analyze the receptor binding characteristics by use of recombinant H                                                                                                         | Swine H1 binds $\alpha$ -2,6 residue better than H1 of A/2009/H1N1 due to difference in positions 200 and 227                                                                                                                                                                      | 149       |
|               | Analyze the receptor binding characteristics by use of recombinant H                                                                                                         | Compared to 1930 H1N1 and PR8, A/2009/H1N1 has stronger binding to $\alpha$ -2,6 sialic acid                                                                                                                                                                                       | 188       |
|               | Analyze the change of amino acid on H                                                                                                                                        | A single point mutation (I219K) in the glycan receptor binding site quantitatively increases its human receptor binding affinity                                                                                                                                                   | 285       |

(Continued on following page)



TABLE 1 (Continued)

| Viral protein | Study design                                                                                                                                                                       | Finding(s)                                                                                                                                                                                                                                                           | Reference |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
|               | Mouse model; evaluate the binding of H from different seasonal H1N1 strains, 1918 H1N1, and A/2009/H1N1, using chimeric viruses                                                    | Viruses expressing A/2009/H1N1 H were associated with significant pathology in the lower respiratory tract and showed low binding activity for surfactant protein D                                                                                                  | 452       |
|               | Analyze the change of amino acid on H                                                                                                                                              | Q223R enhanced infectivity of H pseudotypes in 293T cells                                                                                                                                                                                                            | 535       |
|               | Evaluate the <i>in vitro</i> and <i>in vivo</i> effects of D222G                                                                                                                   | D222G increased virus replication in MDCK cells and pathogenicity in mice                                                                                                                                                                                            | 562       |
|               | Use random peptide library of H from A/2009/H1N1 to identify antigenically important regions                                                                                       | Most antigenic sites are located in the conserved H stem region responsible for membrane fusion                                                                                                                                                                      | 565       |
|               | Mouse model: analyze amino acid changes in mouse-adapted H1N1                                                                                                                      | D131E, S186P, and A198E contribute to virulence in mice                                                                                                                                                                                                              | 572       |
| N             | Use of homology modeling and molecular dynamic stimulation to study the N from different influenza viruses                                                                         | Amino acid residues unique to A/2009/H1N1 are responsible for its affinity for $\alpha$ -2,6 sialic acid                                                                                                                                                             | 291       |
| NS1           | Chickens; evaluate the role of NS1 protein in overcoming innate host defenses and pathogenicity in chickens using recombinant Newcastle disease virus with NS1 gene of A/2009/H1N1 | Recombinant virus expressing the A/2009/H1N1 NS1 grew to higher titers than that expressing H5N1 NS1, antagonized IFN- $\beta$ synthesis more efficiently in HeLa cells than in chicken embryo fibroblast cells, and inhibited PKR activation in infected HeLa cells | 307       |
|               | Mouse and ferret model; compare NS1 protein of A/2009/H1N1 to NS1 of other human-adapted H1N1 viruses                                                                              | Recombinant A/2009/H1N1 expressing NS1 of human-adapted seasonal strains induced less morbidity in mice and reduced titers in upper respiratory tracts of ferrets                                                                                                    | 229       |
|               | Cell culture and mouse model                                                                                                                                                       | XSEV PDZ ligand motif in NS1 contributes to efficient replication of A/2009/H1N1 in cell culture; RSEV, RSKV, and ESEV PDZ motifs in combination with 220W increase mouse pathogenicity                                                                              | 421       |
|               | Effect of NS1 on cellular pre-mRNA polyadenylation and mRNA translation                                                                                                            | NS1 does not contribute to translation of mRNA and does not inhibit cellular pre-mRNA polyadenylation in A549 cells                                                                                                                                                  | 55        |
|               | Vaccine based on NS1                                                                                                                                                               | Live vaccine based on A/2009/H1N1 NS1 can provide protection against infection in mice and ferrets                                                                                                                                                                   | 583       |
| PB1-F2        | Mouse and ferret model; determine the effect of PB1-F2 in A/2009/H1N1 by creating recombinant A/2009/H1N1 expressing PB1-F2                                                        | When comparing wild-type A/2009/H1N1 with recombinant A/2009/H1N1 expressing PB1-F2, there were no significant differences in morbidity and mortality                                                                                                                | 227       |
|               | Cell culture and mouse model; determine the effect of PB1-F2 mutation                                                                                                              | F2-stop12L mutants (a stop-to-leucine substitution at position 12 in the PB1-F2) replicated more efficiently in MDCK cells but did not increase pathogenicity in mice                                                                                                | 421       |
| PA            | Mouse model; analysis of the effect of amino acid substitution                                                                                                                     | L295P responsible for transcription and replication activity                                                                                                                                                                                                         | 271       |
|               | Mouse model                                                                                                                                                                        | PA appears crucial in maintaining viral gene functions (substitution of PA gene impaired activity of all polymerase complexes)                                                                                                                                       | 484       |
|               | <i>In vitro</i> study                                                                                                                                                              | S186 is necessary for the protein to function optimally                                                                                                                                                                                                              | 537       |
|               | Mouse model; analyze amino acid changes in mouse-adapted A/2009/H1N1                                                                                                               | E298K contributes to virulence                                                                                                                                                                                                                                       | 572       |
| NP            | Mouse model; analyze amino acid changes in mouse-adapted A/2009/H1N1                                                                                                               | D101G contributes to virulence                                                                                                                                                                                                                                       | 572       |

<sup>a</sup> H1 numbering is used in denoting polymorphism position. H, hemagglutinin; HTBE, human tracheo-bronchial epithelial cells; ICU, intensive care unit; MDCK, Madin-Darby canine kidney; PKR, protein kinase R.

ing diseases (511). This finding was not unexpected because the A/2009/H1N1 virus is antigenically and structurally similar to the A/1918/H1N1 pandemic virus at antigenic sites Sa, Sb, Ca1, Ca2, and Cb over the globular head of H in crystallization and computer modeling but is different from other H1 viruses (356, 563). However, the preexisting seropositivity rate alone cannot explain some of the observations. For example, a study in Hong Kong showed an absence of preexisting seropositivity in the age group of 61 to 70 years, but the incidence of the A/2009/H1N1 virus was low in this population (579).

### Risk Factors for Severe Disease

In addition to extremes of age and chronic underlying medical illness, other risk factors for severe disease or complications emerged from the A/2009/H1N1 pandemic. Immunosuppressive therapy was reported in a greater percentage of fatal cases in North America than of fatal cases in Asia and the Far East (15.7% versus 6.1%;  $P < 0.001$ ) (127, 152, 170, 181, 187, 305, 313, 365, 435, 574) (Table 4). Additionally, underlying neurological conditions appeared to be more common in fatal cases in North America (37.7% versus 6.7%;  $P < 0.001$ ) (127, 152, 181, 281, 305, 365, 435, 493, 523, 574). Unexpectedly, obesity emerged as a new risk factor for severe influenza, which can predict mortality (364, 458, 522).

Although obesity adversely affects pulmonary function, it has recently been associated with immunodysregulation involving adipokines (269, 298, 407). Pregnancy has been confirmed as a risk factor for severe influenza (363). Pregnant women who were infected during the third trimester were at a higher risk for complications. Notably, no deaths were reported in pregnant women in Asia and the Far East (305). In a mouse model, pregnancy was associated with severe pulmonary damage and cytokine dysregulation, but there was no involvement of the placenta or fetus (85). In addition, smoking and allergies have been proposed as risk factors, but no systematic control studies have been performed (233, 570). In children, neurodevelopmental disorders emerged as a predominant risk factor (8). Additionally, an IgG2 subclass deficiency has been linked to severe pandemic influenza, especially in pregnant women (80, 206), and lower IgG2 levels in severe disease may be associated with cytokine dysregulation (80). However, the usefulness of intravenous immunoglobulin replacement therapy still needs to be confirmed in randomized controlled trials (207).

Primary viral pneumonia is typically seen in patients with underlying comorbidities, such as chronic cardiopulmonary diseases, but 12.5% and 6.3% of cases occur in healthy adults and

TABLE 2 Sequence of important events related to A/2009/H1N1 influenza

| Date              | Important event <sup>a</sup>                                                                                                     |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Mid-February 2009 | Outbreak of respiratory illness in La Gloria, Veracruz, Mexico                                                                   |
| 12 April 2009     | Mexican public health authorities reported outbreak to PAHO                                                                      |
| 15 April 2009     | CDC identified A/2009/H1N1 in a boy from San Diego, CA                                                                           |
| 17 April 2009     | CDC identified A/2009/H1N1 in a girl from Imperial, CA                                                                           |
| 19 April 2009     | Mexico declared a national alert                                                                                                 |
| 21 April 2009     | CDC alerted doctors to a new strain of H1N1 influenza virus                                                                      |
| 24 April 2009     | WHO issued Disease Outbreak Alert                                                                                                |
| 27 April 2009     | WHO raised the pandemic alert from phase 3 to 4                                                                                  |
| 29 April 2009     | WHO raised the pandemic alert from phase 4 to 5                                                                                  |
| 11 June 2009      | WHO raised the pandemic alert from phase 5 to 6                                                                                  |
| 8 July 2009       | Virus strains resistant to oseltamivir identified                                                                                |
| 13 July 2009      | WHO issued recommendations on pandemic H1N1 2009 vaccines                                                                        |
| 5 November 2009   | Detection of infection of farmed swine by the pandemic virus                                                                     |
| 20 November 2009  | Virus mutation detected in fatal and severe cases in Norway                                                                      |
| 2 December 2009   | Oseltamivir-resistant virus identified in hospitalized and immunosuppressed patients                                             |
| 18 February 2010  | WHO issued recommendations for the composition of influenza virus vaccines for the upcoming Northern Hemisphere influenza season |
| 10 August 2010    | WHO issued recommendations for the postpandemic period                                                                           |

<sup>a</sup> CDC, Centers for Disease Control and Prevention; PAHO, Pan American Health Organization; WHO, World Health Organization.

pregnant women, respectively (455). Genetic susceptibility may predispose healthy individuals to severe disease. In a case-control study, single nucleotide polymorphisms (SNPs) in the genes encoding the FcγRIIA protein, the replication protein A-interacting protein, and the complement component 1q subcomponent binding protein and another gene in chromosome 3 were associated with severe pneumonia (591). Interestingly, the SNP in the FCGR2A gene was found more frequently in patients with severe A/2009/H1N1 infection in China; however, this finding was not statistically significant (80). Recently, a genome-wide knockdown study with a short hairpin RNA (shRNA) library identified several important host genes that are involved in the viral life cycle, including fusion, uncoating, transport of the viral RNP complex into and out of the nucleus, replication, transcription and translation of the genome, assembly, and budding (297, 319, 468). The relationship between these host genes and disease severity should be examined further.

HISTOPATHOLOGY AND PATHOGENESIS

A total of 250 autopsies that examined the lungs, livers, spleens, and bone marrow specimens from 2009 pandemic H1N1 cases were reported in the literature (66, 200, 236, 378, 398, 403, 442, 470, 486, 500). Among 190 fatal cases for whom clinical data were reported, the male-to-female ratio was 101:89. The ages ranged from 2 months to 83 years, with a median age of 35 years (66, 200, 236, 378, 398, 403, 470, 486, 500). The interval between symptom onset and death ranged from 1 to 44 days, with a median of 8 days. The histopathological changes in fatal cases depended on the date of death after symptom onset. The changes in patients who died within the first 10 days after symptom onset were dominated by virus-induced cytolysis and acute inflammation, which involved the upper to lower respiratory tracts. The changes in patients who

TABLE 3 Epidemiological characteristics of patients in different continents with laboratory-confirmed A/2009/H1N1 infection

| Epidemiological setting of transmission | Patient characteristics <sup>a</sup> (reference[s])                       |                                                    |                                                      |                                                                 |                                                                            |
|-----------------------------------------|---------------------------------------------------------------------------|----------------------------------------------------|------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------|
|                                         | Asia                                                                      | Oceania                                            | Europe                                               | North America                                                   | South America                                                              |
| Community <sup>b</sup>                  | AR, 5.4–20.6; $R_o$ , 1.22–2.3; $T_g$ , 0.8–1.9 (134, 359, 416, 556, 558) | AR, NM; $R_o$ , 1.16–1.29; $T_g$ , NM (262)        | AR, 6–20; $R_o$ , 0.5; $T_g$ , 2.7 (20, 154, 226)    | AR, — <sup>c</sup> ; $R_o$ , 1.7–1.8; $T_g$ , 2.2–3.3 (65, 543) | AR, 7.7–61; $R_o$ , 1.2–3.1; $T_g$ , 1.9–3.2 (36, 184, 205, 262, 506, 571) |
| Household                               | AR, 8–26.1; $R_o$ , NM; $T_g$ , NM (124, 318)                             | AR, 14.5; $R_o$ , NM; $T_g$ , NM (59)              | NA                                                   | AR, 4–45; $R_o$ , NM; $T_g$ , 3.9 (64, 182, 284, 393, 428, 571) | AR, 35; $R_o$ , 1.8; $T_g$ , 3.61 (436)                                    |
| Nosocomial                              | AR, 23.5; $R_o$ , NM; $T_g$ , NM (10)                                     | NA                                                 | AR, 35; $R_o$ , NM; $T_g$ , NM (114)                 | AR, 5–40; $R_o$ , NM; $T_g$ , NM (74, 172)                      | NA                                                                         |
| School                                  | AR, 21.3; $R_o$ , NM; $T_g$ , NM (326)                                    | NA                                                 | AR, 2–60; $R_o$ , NM; $T_g$ , NM (53, 221, 248, 480) | AR, 2.8–21; $R_o$ , NM; $T_g$ , NM (72, 133, 275, 347, 514)     | NA                                                                         |
| Other                                   | AR, 21–41 <sup>d</sup> ; $R_o$ , NM; $T_g$ , NM (231)                     | AR, 3.5 <sup>e</sup> ; $R_o$ , NM; $T_g$ , NM (18) | NA                                                   | AR, 8–22 <sup>f</sup> ; $R_o$ , NM; $T_g$ , NM (73, 130, 547)   | NA                                                                         |

<sup>a</sup> AR, attack rate (percent);  $R_o$ , basic reproduction number, defined as the number of secondary infections generated by a primary infection in a susceptible population and which thus measures the intrinsic transmissibility of an infectious agent;  $T_g$ , disease generation time, defined as the mean time interval (days) between infection of one person and infection of the people whom individual infects; NA, not available; NM, not mentioned.

<sup>b</sup> In a retrospective analysis of the national surveillance data involving over 2,000,000 cases of A/2009/H1N1 influenza in Japan, it was noted that males <20 years of age may be more likely to suffer from A/2009/H1N1 than females in the same age categories (167).

<sup>c</sup> —, the attack rate in one study was highest among children aged 5 to 14 years (147 per 100,000 population), followed by children aged 0 to 4 years (113 per 100,000). The attack rate for children aged 5 to 14 years was 14 times higher than that for adults aged ≥60 years.

<sup>d</sup> Outbreak among tour group members in China.

<sup>e</sup> Outbreak on a passenger aircraft; their seating was within two rows of infected passengers, implying a risk of infection of about 3.5% for the 57 passengers in those rows.

<sup>f</sup> Including an outbreak on a docked Navy ship in the United States, an outbreak at the U.S. Air Force Academy, and an outbreak on a Peruvian navy ship in the United States.

**TABLE 4** Underlying comorbidities of fatal hospitalized cases with laboratory-confirmed A/2009/H1N1 infection

| Underlying condition          | No. of patients with condition/no. of deaths (% of patients with condition) <sup>a</sup> |                  |                  |                           |                         |
|-------------------------------|------------------------------------------------------------------------------------------|------------------|------------------|---------------------------|-------------------------|
|                               | Asia (n = 952)                                                                           | Africa (n = 104) | Europe (n = 928) | North America (n = 1,200) | South America (n = 156) |
| Chronic respiratory condition | 105/908 (11.6)                                                                           | 11/85 (12.9)     | 181/788 (23)     | 345/1,200 (28.8)          | 37 (23.7)               |
| Chronic cardiac condition     | 46/169 (27.2)                                                                            | 9/71 (12.7)      | 161/928 (17.3)   | 166/751 (22.1)            | 18 (11.5)               |
| Diabetes mellitus             | 33/141 (23.4)                                                                            | 15/85 (17.6)     | 82/620 (13.2)    | 203/1177 (17.2)           | 18 (11.5)               |
| Renal failure                 | 63/908 (6.9)                                                                             | 0                | 35/499 (7.0)     | 71/751 (9.5)              | 8 (5.1)                 |
| Liver cirrhosis               | 42/898 (4.7)                                                                             | 0                | 28/499 (5.6)     | 14/324 (4.3)              | 6 (3.8)                 |
| Rheumatological condition     | 0                                                                                        | 0                | 9/138 (6.5)      | 0                         | 0                       |
| Neurological condition        | 61/916 (6.7)                                                                             | 0                | 127/499 (25.5)   | 292/774 (37.7)            | 0                       |
| Malignancy                    | 9/44 (20.5)                                                                              | 0                | 48/499 (9.6)     | 45/324 (13.9)             | 0                       |
| HIV/AIDS                      | 0                                                                                        | 19/45 (42.2)     | 3/228 (1.3)      | 4/110 (3.6)               | 0                       |
| Immunosuppressive therapy     | 55/898 (6.1)                                                                             | 2/13 (15.4)      | 89/483 (18.4)    | 118/751 (15.7)            | 29 (18.6)               |
| Obesity                       | 103/916 (11.2)                                                                           | 19/86 (22.1)     | 116/928 (12.5)   | 49/517 (9.5)              | 41 (26.8)               |
| Pregnancy                     | 0                                                                                        | 29/101 (28.7)    | 19/928 (2)       | 25/447 (5.6)              | 10 (6.4)                |
| No risk factor                | 46/169 (27.2)                                                                            | 16/76 (21.1)     | 85/259 (29.4)    | 6/19 (31.6)               | 51 (32.7)               |

<sup>a</sup> The information should be interpreted with care because the denominators of the underlying conditions are different. References are as follows: for patients in Asia, references 131, 302, 305, 493, and 574; for patients in Africa, references 15 and 313; for patients in Europe, references 67, 152, 187, 435, 523, and 544; for patients in North America, references 68, 75, 127, 181, 281, and 365; and for patients in South America, reference 170.

died much later with intensive care support were dominated by both damage and reparative processes.

### Histopathological Changes in the Respiratory Tract

Upper and middle airway infections caused by the A/2009/H1N1 virus were characterized by the multifocal destruction and desquamation of the pseudocolumnar and columnar epithelia, with significant submucosal edema and hyperemia to the extent of thrombus formation at the bronchiolar level. The acute inflammation could be severe, as evidenced by hemorrhagic tracheo-bronchitis and desquamative bronchiolitis with necrosis of the bronchiolar wall. Once necrosis occurred, there was infiltration by polymorphs and mononuclear cells. The histological changes in influenza pneumonia included interstitial edema with inflammatory infiltrate, alveolar proteinaceous exudation with membrane formation, capillary thrombosis, necrosis of the alveolar septa, intra-alveolar hemorrhage, desquamated pneumocytes with pyknotic nuclei into the alveolar spaces, and diffuse alveolar damage with dominant mononuclear interstitial infiltration by lymphocytes and histiocytes (Fig. 4A and B) (200, 378, 470). During the late stage, organizing diffuse alveolar damage, fibrosis, type II pneumocyte hyperplasia, epithelial regeneration, and squamous metaplasia were found, which are compatible with the fibroproliferative phase of acute respiratory distress syndrome (ARDS) and diffuse alveolar damage (Fig. 4C). Histochemical staining for the viral NP showed that type II pneumocytes were the predominantly affected cell type (470), and the staining was positive in patients who died within 3 days after symptom onset. Bacterial coinfections were documented in 26% to 33.3% of the case series that were analyzed (Fig. 4D) (66, 200, 236, 378, 470). The most common bacteria found in the autopsy series included *Streptococcus pneumoniae*, *Streptococcus pyogenes*, *Staphylococcus aureus*, community-associated methicillin-resistant *S. aureus*, *Streptococcus mitis*, *Haemophilus influenzae*, and *Acinetobacter baumannii* (66, 107, 236). Mouse studies suggested that during A/2009/H1N1 infection, the innate defense against secondary pneumococcal infection was impaired due to the increased gamma interferon (IFN- $\gamma$ ) response during the recovery stage of infection (496).

### Histopathological Changes in Extrapulmonary Sites

In addition to the pulmonary pathologies, the autopsy studies revealed pathological changes in other sites. Myocarditis with myofibril degeneration and interstitial lymphocytic infiltrate were found in several fatal cases of 2009 pandemic H1N1 infection (Fig. 4E) (195, 510). Hemophagocytosis was found in several severe cases, as in cases of influenza A H5N1 virus infection (236, 470, 510). Pulmonary thromboembolism was evident in 32 cases (Fig. 4F and 4G) (200, 236, 378, 470, 510). Splenic infarction associated with thrombosed arterial supply was observed in the postmortem examinations (Fig. 4H).

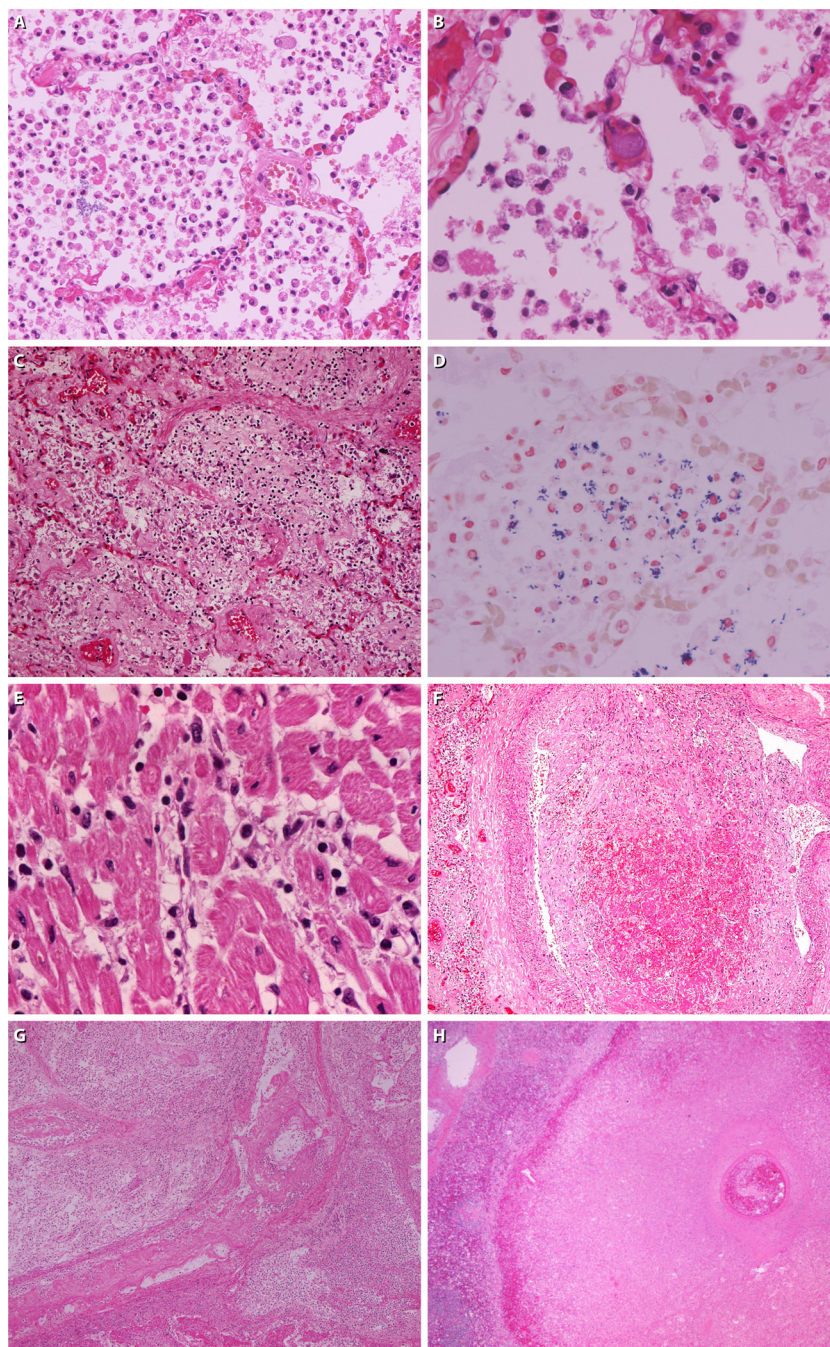
### Host Cytokine Profile

The majority of the studies on the blood cytokine and chemokine levels showed that interleukin-6 (IL-6), IL-8, IL-10, IL-15, IFN- $\gamma$ , and tumor necrosis factor alpha (TNF- $\alpha$ ) were consistently elevated in severe disease (Table 5). Although the blood IL-17 level was found to be elevated in one study (31), this finding was not consistent and could be related to the sampling times at the different stages of the disease process (13, 225, 308, 510). IL-17 has recently been shown to be important in the recruitment of B cells (536). *In vitro* studies of the mRNA expression profiles and cytokine activation levels in virus-infected peripheral blood mononuclear cells showed no significant differences between the A/2009/H1N1 and seasonal H1N1 viruses except for the downregulation of zinc finger proteins and small nucleolar RNAs by seasonal H1N1 virus (341). The cytokine activation findings in animal challenge experiments with mice, ferrets, and macaque monkeys were heterogeneous, as expected with species variations (28, 85, 274, 294, 461).

### Host Immune Response

Upon infection, influenza virus will trigger a series of host immune responses. The initial innate immune response includes the following: the expression of pattern recognition receptors (PRRs), such as Toll-like receptors (150), the retinoic acid-inducible gene I (RIG-I) protein (443), and NOD-like receptors; the complement





**FIG 4** Histopathological examination in fatal cases of A/2009/H1N1. Hematoxylin and eosin (H&E) staining was used for panels A to C and E to H; Gram staining was used for panel D. (A) Lung parenchyma showing the acute phase of viral pneumonia and ARDS, with numerous macrophages within alveolar space. The alveolar septa are congested. Magnification,  $\times 200$ . (B) A pneumocyte displaying cytopathic change with an enlarged nucleus during the acute phase of ARDS due to viral pneumonia. Magnification,  $\times 400$ . (C) Lung parenchyma showing the chronic fibroproliferative phase of diffuse alveolar damage. The alveolar septa are thickened, and the alveolar spaces are replaced by fibrogranulation tissue. Magnification,  $\times 200$ . (D) The alveolar spaces contain many Gram-positive cocci, with some being ingested by macrophages in a patient with secondary bacterial pneumonia. Magnification,  $\times 200$ . (E) A case with myocarditis, showing lymphoid infiltrate in the myocardium. Magnification,  $\times 200$ . (F) Organized thrombus of a branch of pulmonary artery. Magnification,  $\times 100$ . (G) A branch of pulmonary artery with recent thrombus formation. The surrounding lung parenchyma shows heavy acute inflammatory infiltration. Magnification,  $\times 40$ . (H) Splenic infarct associated with a thrombosed arteriole. Magnification,  $\times 40$ . (All photos courtesy of Chung-Ying Leung, reproduced with permission.)

cascade; and antimicrobial peptides. The subsequent adaptive immune response consists of B and T lymphocytes, which are triggered by antigen-presenting cells, such as dendritic cells and macrophages.

Studies with prototype strains of the influenza A H1N1 virus showed that the virus triggers an immediate response from the innate immune system through nonspecific PRRs, including TLR3, -7, and -9 (150, 220), at the plasma membrane or within



**TABLE 5** Changes in blood levels of cytokines and chemokines in severe and mild cases of A/2009/H1N1 infection

| Cytokine <sup>a</sup> | Reference(s) in which level was found to be: |                       |                                        |
|-----------------------|----------------------------------------------|-----------------------|----------------------------------------|
|                       | Higher in severe cases                       | Lower in severe cases | Not different in severe and mild cases |
| G-CSF                 | 510                                          |                       |                                        |
| GM-CSF                | 30                                           |                       | 510                                    |
| VEGF                  | 30                                           |                       |                                        |
| IFN- $\alpha$ 2       | 510                                          |                       |                                        |
| IFN- $\alpha$         | 3                                            |                       |                                        |
| IFN- $\gamma$         | 30                                           |                       | 225, 308, 510, 541                     |
| IL-1 $\alpha$         | 510                                          |                       | 13                                     |
| IL-1RA                | 13, 30                                       |                       |                                        |
| IL-2                  | 13                                           |                       | 510                                    |
| IL-5                  | 498 <sup>b</sup>                             |                       | 510                                    |
| IL-6                  | 13, 30, 225, 308, 340, 510                   |                       | 3, 541                                 |
| IL-8 (CXCL-8)         | 30, 225, 340, 510                            |                       | 3, 13, 541                             |
| IL-10                 | 13, 30, 510                                  |                       | 3                                      |
| IL-12p70              | 30                                           |                       | 13, 510                                |
| IL-15                 | 225, 510                                     |                       |                                        |
| IL-17                 | 31                                           | 510 <sup>c</sup>      | 13, 225, 308                           |
| IP-10                 | 30, 308, 510                                 |                       | 31, 225                                |
| MCP-1 (CCL2)          | 30, 340, 510                                 | 3                     | 31                                     |
| MIP-1 $\alpha$ (CCL3) | 13                                           |                       | 510                                    |
| MIP-1 $\beta$ (CCL4)  | 13                                           |                       | 31, 510                                |
| TNF- $\alpha$         | 31, 225, 510                                 |                       | 13                                     |
| sTNFR1                | 340                                          |                       |                                        |
| TGF- $\beta$ 1        | 541                                          |                       |                                        |

<sup>a</sup> G-CSF, granulocyte colony-stimulating factor; GM-CSF, granulocyte-macrophage colony-stimulating factor; VEGF, vascular endothelial growth factor; IL-1RA, IL-1 receptor antagonist; IP-10, IFN- $\gamma$ -induced protein 10; MCP-1, monocyte chemoattractant protein 1; MIP-1 $\beta$ , macrophage inflammatory protein 1 $\beta$ ; sTNFR1, soluble TNF receptor 1; TGF- $\beta$ 1, transforming growth factor  $\beta$ 1.

<sup>b</sup> Compared children with or without pneumonia.

<sup>c</sup> Lower in severe cases only if  $\leq 3$  days after onset of symptoms.

endosomes. TLR3 recognizes double-stranded RNA (dsRNA); however, it is unclear whether the influenza virus produces dsRNA in infected cells. TLR7 and TLR8 recognize uridine-rich sequences of single-stranded RNA (ssRNA). The triggering of these TLRs will lead to the recruitment of MyD88 and the activation of transcription factor NF- $\kappa$ B and mitogen-activated protein kinase (MAPK) inflammatory cascades. After virus entry and uncoating, viral RNA is recognized by the cytoplasmic RIG-I-like receptors, and the melanoma differentiation-associated gene 5 (MDA5), which are regulated by LPG2 and ubiquitin ligase (299). A dsRNA binding protein, called PACT, is a potent binder and activator of RIG-I and boosts the antiviral interferon response (316, 352). The activation of RIG-I results in a type I interferon response in epithelial cells. However, one key virulence factor of influenza virus, NS1, can inhibit this RIG-I-dependent interferon production in two ways. First, NS1 can interfere with the activation of RIG-I by viral ssRNA harboring free 5'-triphosphate groups (443). Alternatively, NS1 reduces RIG-I signal transduction by inhibiting the TRIM25-mediated RIG-I-CARD ubiquitination (190). The significance of these molecular interactions between the viral and host factors is shown in cell culture and mouse challenge models infected with the prototype influenza A H1N1

virus (PR8). Compared with the PR8 virus with NS1 deleted, the cells and mice infected by the wild-type virus do not mount an innate immune response for almost 2 days postinfection as indicated by cytokine and chemokine assays and histopathology, during which the viral load starts to peak. This quiet stealth phase was followed by a sudden burst of lung inflammation in mice, with the subsequent initiation of adaptive immunity by migratory lung dendritic cells (390). These findings are compatible with the clinical findings that the nasopharyngeal viral load in infected human volunteers had already peaked by the time the patients had symptoms (244). It is unknown whether severe or fatal cases had a higher peak viral load because the majority of them did not present early; however, viral load monitoring after patient admission showed that patients had a delayed clearance of viral load in respiratory specimens, with concomitantly higher serum proinflammatory cytokines and chemokines (510). Recently, it has also been shown that agonism of sphingosine-1-phosphate signaling in the endothelium could suppress the influenza virus-induced cytokine storm, suggesting the role of endothelial cells in the modulation of inflammation triggered by influenza virus (504). It is possible that the final clinical outcome of the host was determined by the balance between the amount of tissue damage due to virus-infected cells, which is governed by the innate immune control and the subsequent adaptive immune control, the inflammatory damage inflicted by the innate immune response and the subsequent adaptive immune response, the pulmonary reserve, and regenerative power of the patient.

Infection by the A/2009/H1N1 virus may sometimes induce broadly reactive neutralizing antibodies that target the H stalk and globular head domain in multiple influenza virus strains. In addition, A/2009/H1N1 infection induced a cytotoxic T lymphocyte response that targeted internal type-specific viral core proteins (211, 555). However, patients with severe influenza were shown to have lower serum complement levels and higher titers of antibodies, with a lower avidity for attaching to H, than those with mild cases (391). The immune complex formed between low-avidity antibodies and viral antigens in infected cells fixed complement, which activated the inflammatory cascade and led to the exaggerated inflammation of infected lung tissue. This finding revealed a new phenomenon of pathogenesis for pandemic A/2009/H1N1 influenza virus (391), which requires further confirmation.

Many studies have been conducted on the cross-reactive T lymphocyte-mediated immunity directed against both the A/2009/H1N1 and seasonal H1N1 viruses (Table 6). The amount of cross-reactivity varied from 18.1% to 69% and was directed mainly against the internal proteins, such as M1, NP, and PB1 (213). The prevalence of cross-reactive antibodies against the A/2009/H1N1 and seasonal H1N1 viruses in the sera of healthy individuals younger than 65 years of age was generally less than 6% before 2009 (77). When these healthy individuals were immunized with the seasonal H1N1 vaccine, up to 22% of these vaccinees had a protective neutralizing antibody titer (232). When mice were immunized with the seasonal H1N1 live attenuated vaccine, decreased morbidity and mortality were observed (92). This protection was largely due to cell-mediated immunity.

Using flow cytometry, increased regulatory T (Treg) lymphocyte activity was observed in peripheral blood samples from infected versus noninfected patients at 1 to 2 days after symptom onset, but the increase was more significant in moderate cases than in severe cases (222). In severe cases, B lym-

**TABLE 6** Cross-reactive adaptive immune responses against A/2009/H1N1 virus in humans and experimental animals<sup>a</sup>

| Type of response | Study design (location)                                                                                                                                                                                                                   | Subjects tested                                                                                                                                                              | Findings                                                                                                                                                                                                                                                                                                                                                                                                                                    | Reference |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| T cell           | Measure the extent of cross-reactivity of seasonal H1N1 influenza A virus-specific CD4 T cells with A/2009/H1N1 epitopes (USA)                                                                                                            | 11 persons (age NM)                                                                                                                                                          | 15 A/2009/H1N1 peptides were found to cross-react with seasonal influenza virus-specific T cells                                                                                                                                                                                                                                                                                                                                            | 196       |
|                  | Compare the T cell responses in patients with or without A/2009/H1N1 infection (Australia)                                                                                                                                                | 6 healthy donors and 2 patients with laboratory-confirmed A/2009/H1N1 infection                                                                                              | A/2009/H1N1 uninfected patients showed no preexisting T cell response to the 2009-NP418 variant or the 1918-NP418 variant; natural infection with the A/2009/H1N1, however, elicited CD8 <sup>+</sup> T cells specific for the 2009-NP418 and 1918-NP418 epitopes                                                                                                                                                                           | 211       |
|                  | Measure the cross-reactivity of memory T cell immunity against A/2009/H1N1 in PBMCs collected before the 2009 pandemic (USA)                                                                                                              | 20 adults                                                                                                                                                                    | 69% of the epitopes recognized by CD8 <sup>+</sup> T cells were conserved among the seasonal H1N1 influenza virus and A/2009/H1N1; cross-reactive memory T cell immunity was present in the general population                                                                                                                                                                                                                              | 213       |
|                  | Measure the T cell responses in a cohort of healthy nonimmunized adults; the blood samples were collected before summer 2009 (USA)                                                                                                        | 46 healthy nonimmunized adults                                                                                                                                               | 5/46 subjects had T cell response <30% of coexistent seasonal response; 19/46 subjects had T cell response greater than seasonal response                                                                                                                                                                                                                                                                                                   | 491       |
|                  | Measure the CTL response to A/2009/H1N1 in patients without prior infection with A/2009/H1N1 (Hong Kong)                                                                                                                                  | 12 adults                                                                                                                                                                    | CTL from uninfected individuals could directly lyse A/2009/H1N1-infected target cells and produce IFN- $\gamma$ and TNF- $\alpha$ ; 17/94 (18.1%) of influenza A virus CD8 T-cell epitopes were conserved in A/2009/H1N1, and >1/2 of these conserved epitopes were derived from M1 protein; seasonal influenza vaccination could expand the functional M1 <sub>58-66</sub> epitope-specific CTLs in 20% of HLA-A2 <sup>+</sup> individuals | 518       |
|                  | Investigate the degree of T-cell cross-reactivity between seasonal influenza A (sH1N1, H3N2) from 1968 to 2009 and A/2009/H1N1 strains (Canada)                                                                                           | NA                                                                                                                                                                           | T cell cross-reactivity was estimated to be 52%, and maximum conservancy was found between sH1N1 and A/2009/H1N1 with a significant correlation                                                                                                                                                                                                                                                                                             | 162       |
|                  | Assess the cross-reactive T cell immunity in mice (the Netherlands)                                                                                                                                                                       | Mice                                                                                                                                                                         | Mice with prior H3N2 infection displayed reduced weight loss after challenge infection and cleared the A/2009/H1N1 more rapidly; virus-specific CD8 <sup>+</sup> T cells in concert with CD4 <sup>+</sup> T cells were responsible for the observed protection (by adoptive transfer experiments)                                                                                                                                           | 258       |
|                  | Evaluate the memory T cell repertoire in healthy adults (USA)                                                                                                                                                                             | 9 healthy adults aged 18–50 yr                                                                                                                                               | Most individuals had abundant circulating CD4 T cells that recognized influenza virus-encoded proteins                                                                                                                                                                                                                                                                                                                                      | 457       |
|                  | Examine the production of cytokines in response to virus from CD8 <sup>+</sup> T cells from subjects who had no evidence of exposure to A/2009/H1N1 and had blood collected prior to the emergence of the pandemic in April of 2009 (USA) | 9 healthy adults aged 18–49 yr                                                                                                                                               | Most subjects exhibited cytokine positive CD8 <sup>+</sup> T cells in response to A/2009/H1N1                                                                                                                                                                                                                                                                                                                                               | 464       |
|                  | Compare the T cell and antibody responses elicited by trivalent inactivated influenza vaccine and live attenuated influenza vaccine (USA)                                                                                                 | 30 healthy adults                                                                                                                                                            | Both vaccines boosted preexisting T cells to the seasonal and pandemic H, but responses were significantly greater following immunization with LAIV; antibody titers were significantly boosted only by TIV                                                                                                                                                                                                                                 | 492       |
| Antibody         | Measure cross-reactive antibody response to A/2009/H1N1 before and after seasonal influenza vaccination (2005 to 2009) (USA)                                                                                                              | Age 6 mo to 9 yr ( <i>n</i> = 28), 18–59 yr ( <i>n</i> = 30), >60 years ( <i>n</i> = 42)                                                                                     | Before seasonal influenza vaccination (% with VN titer >160): children, no cross-reactive antibody; 18–64 yr, 6–9%; >60 yr, 33%; after seasonal influenza vaccine (increase in cross-reactive antibody titer): children, no increase; 18–64 yr, 2-fold; >60 yr, no increase in cross-reactive antibody                                                                                                                                      | 77        |
|                  | Measure the preexisting cross-reactive antibody response to A/2009/H1N1 before and after seasonal influenza vaccination (USA)                                                                                                             | Born before 1950 ( <i>n</i> = 115), born after 1980 ( <i>n</i> = 107), age 6 mo to 9 yr ( <i>n</i> = 55), age 18–64 yr ( <i>n</i> = 231), age $\geq$ 60 yr ( <i>n</i> = 113) | Proportion of individuals with cross-reactive antibody (VN): born before 1950, 34% (neutralizing antibody titer, $\geq$ 80); born after 1980, 4% (neutralizing antibody titer, $\geq$ 40). Increase in cross-reactive antibody to 2009 H1N1 by 4-fold after seasonal influenza vaccination (VN): 6 mo to 9 yr, 0%; 18–64 yr, 12–22%; >60 yr, <5%                                                                                            | 232       |
|                  | Measure the cross-reactive antibody levels against A/2009/H1N1 in sera collected between 2004 and 2005 by HI assay (Finland)                                                                                                              | Born between 1909 and 1919 ( <i>n</i> = 27), between 1920 and 1929 ( <i>n</i> = 104), between 1930 and 1939 ( <i>n</i> = 125), 1940 or after ( <i>n</i> = 775)               | Proportion of patients with cross-reactive antibody (HI titer, > 10): born between 1909 and 1919, 96%; born between 1920 and 1929, 56.7%; born between 1930 and 1939, 13.6%; born after 1939, <10%                                                                                                                                                                                                                                          | 270       |
|                  | Measure the antibody titer (by HI and VN assay) in sera collected in 2008 (France)                                                                                                                                                        | 100 adults                                                                                                                                                                   | Seasonal 2007 H1N1 infection was an independent predictor of elevated preexposure antibody titers against A/2009/H1N1                                                                                                                                                                                                                                                                                                                       | 345       |
|                  | Study neutralization response to A/2009/H1N1 in patients with prior immunization with 1976 “swine flu” vaccine (USA)                                                                                                                      | Age $\geq$ 55 yr ( <i>n</i> = 116), received “swine flu” vaccination in 1976 ( <i>n</i> = 46), age 0–18 yr ( <i>n</i> = 20)                                                  | Receipt of 1976 “swine flu” vaccine enhanced neutralization response to A/2009/H1N1 (VN titers of $\geq$ 160: vaccine recipients, 17.4%; vs non-vaccine recipients, 4.3% [ <i>P</i> = 0.018])                                                                                                                                                                                                                                               | 379       |
|                  | Animal model (mice, ferrets, minipigs); study the cross-reactive immunity of seasonal influenza vaccine (SW/Korea/CAN01/04; A/Brisbane/59/07) on A/2009/H1N1 (South Korea)                                                                | NA                                                                                                                                                                           | Although receipt of SW/Korea/CAN01/04 induced detectable cross-reactive antibody against A/2009/H1N1, active virus replication and virus shedding were not suppressed                                                                                                                                                                                                                                                                       | 431       |
|                  |                                                                                                                                                                                                                                           |                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                             |           |

(Continued on following page)

TABLE 6 (Continued)

| Type of response               | Study design (location)                                                                                                                                               | Subjects tested                                                                                                                            | Findings                                                                                                                                                                                                                                                                                        | Reference |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
|                                | Measure cross-reactive antibody response to A/2009/H1N1 in serum samples collected in 2004 by HI, SRH, and VN assay (Italy)                                           | Born between 1909 and 1938 ( $n = 201$ ), 1939 and 1948 ( $n = 193$ ), 1949 and 2004 ( $n = 193$ )                                         | Proportion of patients with HI titer of $\geq 40$ : born between 1909 and 1938, 22.4%; born between 1939 and 1948, 12.4%; born between 1949 and 2004, 6.7%                                                                                                                                      | 459       |
|                                | Mouse model; expose mice to virus (1947 virus A/FM/1/47 or 1934 virus A/PR/8/34) and measure the cross-protective immune responses to mouse-adapted A/2009/H1N1 (USA) | NA                                                                                                                                         | Mice exposed to 1934 and 1947 H1N1 viruses were protected against lethal challenge with mouse-adapted A/2009/H1N1                                                                                                                                                                               | 477       |
|                                | Measure the cross-reactive antibody titers in healthy volunteers mostly born before 1958 (Singapore)                                                                  | 50 adults                                                                                                                                  | HI or VN titer of $\geq 40$ : 0% among those 40–80 yr old                                                                                                                                                                                                                                       | 502       |
|                                | Measure the cross-reactive antibody titer (China)                                                                                                                     | Age 7 to 84 yr ( $n = 4,043$ )                                                                                                             | HI titer of $\geq 40$ , 1.7%; VN titer of $\geq 40$ , 0.3%                                                                                                                                                                                                                                      | 95        |
|                                | Measure the cross-reactive antibody titers in sera collected in the prepandemic and early epidemic phases before widespread community transmission (Singapore)        | 838 members of the community, 1,213 military personnel, 558 hospital staff, 300 from long-term care facilities                             | HI titer of $\geq 40$ , 2.6% of community members, 9.4% of military personnel, 6.6% of hospital staff, 6.7% of subjects from long-term care facilities                                                                                                                                          | 101       |
|                                | Measure the VN titer in blood samples collected from donors in 1999 and 2009 (Japan)                                                                                  | 315 persons                                                                                                                                | Most patients born after 1920 had no neutralizing antibody against A/2009/H1N1                                                                                                                                                                                                                  | 274       |
|                                | Measure the background prepandemic cross-reacting antibodies to the A/2009/H1N1 in older populations (Australia)                                                      | 259 serum samples from persons aged 60 yr or older                                                                                         | HI titer of $\geq 40$ : $\geq 60$ yr, 37.5%; $\geq 85$ yr, 60%; 60–64 yr, 12%                                                                                                                                                                                                                   | 39        |
|                                | Measure the cross-reactive antibody titer (Taiwan)                                                                                                                    | 229 stored sera from donors born between 1917 and 2008                                                                                     | Neutralizing titer of $\geq 160$ : $>80$ yr, 59%. Neutralizing titer of $\geq 40$ : children 6 mo to 9 yr, 4% (only those children with neutralizing titer to pandemic H1N1 virus had high neutralizing titers to seasonal H1N1 virus)                                                          | 110       |
|                                | Measure the cross-reactive antibody titer (France)                                                                                                                    | 1,693 serum samples collected in 2007 to 2008                                                                                              | HI titer of $\geq 40$ increased from 40.5% in those 0–24 yr to 70% in elderly                                                                                                                                                                                                                   | 143       |
|                                | Measure the cross-reactive antibody in IVIG preparation (USA)                                                                                                         | 6 IVIG samples prepared in prepandemic period, sera from 19 Kawasaki patients treated with IVIG                                            | All the IVIG preparations had significant levels of cross-reactive specific antibody; 18 out of 19 Kawasaki patients treated with IVIG had significant increase in cross-reactive specific antibody                                                                                             | 261       |
| Clinical                       | Ferret model; effect of prior infection with seasonal influenza A virus on the outcome of A/2009/H1N1 challenge (Australia)                                           | NA                                                                                                                                         | Prior infections with seasonal influenza A viruses reduced the incidence of infection, amount and duration of virus shedding, and frequency of transmission following A/2009/H1N1 challenge                                                                                                     | 335       |
| Vaccine                        | Measure the neutralizing antibody titers before and after immunization with 2009 Southern Hemisphere seasonal influenza vaccine (Singapore)                           | Age 19–46 yr ( $n = 51$ )                                                                                                                  | Postvaccination, 12% had 4-fold rise in VN titer to A/2009/H1N1                                                                                                                                                                                                                                 | 343       |
|                                | Mouse model; protection against lethal challenge with A/2009/H1N1 by 1918-like and classical swine H1N1-based vaccine (USA)                                           | NA                                                                                                                                         | Immunization with 1918-like or classical swine H1N1 vaccine completely protected C57B/6 mice from lethal challenge with A/2009/H1N1                                                                                                                                                             | 373       |
|                                | Mouse model; evaluate the cross-protection induced by immunization with 2009 pH1N1 vaccines following a lethal challenge with 1918 H1N1 (USA)                         | NA                                                                                                                                         | Vaccination with A/2009/H1N1 protected mice from lethal challenge with 1918 virus                                                                                                                                                                                                               | 164       |
| Vaccine/prior infection        | Mouse model; evaluate the effect of prior seasonal H1N1 infection and seasonal influenza vaccine on the immune response to pandemic influenza vaccine/infection (USA) | NA                                                                                                                                         | p-LAIV induced a cellular response and nonneutralizing antibody production but only partial protection from A/2009/H1N1 challenge; primary infection with sH1N1 followed by p-LAIV resulted in cross-reactive antibody and robust cellular response and was associated with complete protection | 92        |
|                                | Ferret model; evaluate the effect of prior seasonal H1N1 infection or vaccine (USA)                                                                                   | NA                                                                                                                                         | Seasonal influenza vaccine was unable to alter subsequent morbidity or transmission in ferrets                                                                                                                                                                                                  | 434       |
| Vaccine                        | Evaluate the antibody titer to A/2009/H1N1 after seasonal influenza vaccine (Australia)                                                                               | 20 children with median age of 4 yr                                                                                                        | Only 2 children were seropositive (HI titer = 40) for A/2009/H1N1                                                                                                                                                                                                                               | 380       |
|                                | Evaluate the effect of prior seasonal influenza vaccine (USA)                                                                                                         | 30 healthy subjects between age 18 and 49 yr who received vaccine between October and November 2007 (15 received LAIV and 15 received TIV) | Only preexisting T cells, and not antibody titer, to pandemic H1 boosted after LAIV                                                                                                                                                                                                             | 492       |
|                                | Mouse model; evaluate the effect of prior live seasonal influenza vaccine (USA)                                                                                       | NA                                                                                                                                         | Mice immunized with seasonal LAIV had decreased morbidity and mortality from A/2009/H1N1 infection<br>Protective immunity primarily dependent upon CD4 <sup>+</sup> T cells but not CD8 <sup>+</sup> T cells                                                                                    | 496       |
|                                | Mouse model; evaluate the effect of a vaccine using synthetic consensus H1 antigen (USA)                                                                              | NA                                                                                                                                         | 80% of mice challenged with A/2009/H1N1 were protected                                                                                                                                                                                                                                          | 539       |
|                                | Evaluate the cross-reactivity after inactivated seasonal influenza vaccine (USA)                                                                                      | 120 adults aged 20–64 yr<br>59 elderly aged 65 yr or above                                                                                 | 22% of adults and 34% of elderly showed a $\geq 4$ -fold increase in HI titers after TIV vaccination                                                                                                                                                                                            | 561       |
| Prior seasonal virus infection | Mouse model; evaluate the effect of prior seasonal influenza virus infection on subsequent pH1N1 infection (USA)                                                      | NA                                                                                                                                         | Sequential infection with viral strains with different surface glycosylation can prime the host for immunopathology if a neutralizing antibody response matching the T cell response is not present                                                                                             | 538       |

<sup>a</sup> HI, hemagglutination inhibition; CTL, cytotoxic T lymphocytes; IFN, interferon; IVIG, intravenous immunoglobulin; LAIV, live attenuated influenza vaccine; NA, not applicable; PBMCs, peripheral blood mononuclear cells; p-LAIV, pandemic live attenuated influenza vaccine; SRH, single-radial-hemolysis; TIV, trivalent influenza vaccine; TNF, tumor necrosis factor; VN, virus neutralization.



TABLE 7 Adaptive immune response in patients infected by A/2009/H1N1 virus<sup>a</sup>

| Cell type         | Study design                                                                                                                             | Results                                                                                                                                                                                                                                                                                                                 | Reference |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| T cell            | Compare T cell function in 6 severe and 22 mild cases                                                                                    | T cells from severe cases had impaired effector cell differentiation and failed to respond to mitogenic stimulation; massive expression of CD95 marker found on anergic T cells, suggesting apoptosis-related mechanism                                                                                                 | 3         |
| T cell and B cell | Compare T cells in 31 A/2009/H1N1 cases, 18 patients with flu-like illness, and 10 healthy volunteers                                    | A/2009/H1N1-infected patients had reduced numbers of CD4 lymphocytes and B lymphocytes but increased numbers of T-regulatory lymphocytes                                                                                                                                                                                | 198       |
| T cell            | Compare T cells in 53 A/2009/H1N1 cases and 21 healthy controls                                                                          | A/2009/H1N1-infected patients had T cell activation and preferential loss of Th17 subset at the early stage of infection; the functional loss of Th17 cells is likely due to upregulated IFN- $\alpha$                                                                                                                  | 288       |
|                   | Compare T cells in 36 severe cases, 40 moderate cases, and 20 healthy volunteers                                                         | At 1-2 days from onset, the frequency of Treg cells was higher in moderate than in severe cases                                                                                                                                                                                                                         | 222       |
|                   | Compare T cells in 9 severe and 7 mild cases                                                                                             | Pretreatment and posttreatment CD4 <sup>+</sup> and CD8 <sup>+</sup> T cell counts, measured by flow cytometry, did not differ significantly between the groups ( $P > 0.05$ ); analysis by the paired-sample $t$ test showed a significant increase in posttreatment CD4 <sup>+</sup> T cells in the severe cases only | 541       |
| B cell            | Compare B cells in 36 severe cases, 40 moderate cases, and 20 healthy volunteers                                                         | B cells were increased in severe cases                                                                                                                                                                                                                                                                                  | 222       |
| NK cell           | Comparison between A/2009/H1N1, H5N1, and 1918 H1N1, using pseudotyped particles                                                         | Much stronger NK activation was triggered by H5N1 and 1918 H1N1 than by A/2009/H1N1                                                                                                                                                                                                                                     | 158       |
|                   | Analyze the interaction of NK-activating receptors NKp46 and A/2009/H1N1 or H5N1 in both <i>in vitro</i> and mouse models                | NKp46 binds to H of A/2009/H1N1, leading to killing both <i>in vitro</i> and in the mouse model; NKp46-H5N1 interactions cannot elicit direct killing of infected cells                                                                                                                                                 | 1         |
|                   | Describe the cellular immunology profile in 3 patients with rapidly progressive infection compared with 7 healthy uninfected individuals | NK cells were markedly reduced in 3 patients with progressive infection                                                                                                                                                                                                                                                 | 148       |
|                   | Compare NK cells in 36 severe cases, 40 moderate cases, and 20 healthy volunteers                                                        | Frequency of NK cells was lowest in severe cases                                                                                                                                                                                                                                                                        | 222       |

<sup>a</sup> H, hemagglutinin; IFN, interferon; NK, natural killer; Th, T helper; Treg, regulatory T cell.

phocytes were increased, T cells had impaired effector cell differentiation and proliferation, and there were fewer NK cells (Table 7).

## CLINICAL MANIFESTATIONS

Similar to the case for seasonal human influenza, which is a common cause of acute upper and lower respiratory tract infections in community and health care settings, A/2009/H1N1 disease ranges from a febrile upper respiratory tract infection to fulminant primary viral pneumonia or secondary bacterial pneumonia with ARDS, requiring intensive care and respiratory support. A study demonstrated that patients who subsequently developed ARDS and succumbed had a slower decline in nasopharyngeal viral loads, had higher plasma levels of proinflammatory cytokines and chemokines, and were more likely to have viremia, bacterial coinfections, or a complication of myocarditis than patients who had mild disease or who survived with ARDS (510). Other studies demonstrated that pneumococcal coinfection played a similar role in causing mortality (200, 378, 423) as it did during the Spanish flu (112). The clinical features of A/2009/H1N1 influenza virus infection were similar to those of seasonal influenza (Table 8). However, there were notable differences between the fatal cases reported in North America and those in Asia (Table 4). The incidences of diarrhea (23.7% versus 4.3%;  $P < 0.001$ ) and myalgia (40.9% versus 13.3%;  $P < 0.001$ ) were significantly higher in North American patients. Regarding hospitalized patients, significantly higher rates of admission to intensive care units (ICUs) ( $P < 0.001$ ), mechanical ventilation ( $P < 0.001$ ), and death ( $P < 0.001$ ) were ob-

served in North America than in Asia (chi-square test) (Table 9) (32, 69, 75, 131, 281, 322, 365, 418, 493, 511). This difference may be due to the different thresholds of hospitalization in the different cultures. However, the lower overall clinical manifestation of diarrhea and myalgia in Asians also suggests that Asians may have a lower rate of severe disease.

In addition to upper and lower respiratory tract diseases, extrapulmonary manifestations were also reported for A/2009/H1N1 infection (Table 10) (42, 49, 61, 224, 323, 370, 376, 419, 485, 510). These manifestations included hepatitis, myocarditis, rhabdomyolysis, renal failure, systemic or pulmonary vascular thrombosis, reactive hemophagocytosis, and, in children, acute necrotizing encephalopathy or encephalopathy. Renal failure was associated with an increased risk of death. Several of these manifestations were similar to those found in patients with H5N1 infection. Influenza-associated encephalopathy has been associated with thermolabile carnitine palmitoyltransferase II variants. The disease can be rapidly fatal within 1 to 4 days of acquisition (370).

## CLINICAL MANAGEMENT AND LABORATORY DIAGNOSIS

There are no pathognomonic signs and symptoms of influenza virus infections (169, 511). The clinical, routine laboratory and radiologic findings are not distinguishable from those associated with other causes of influenza-like illness, severe community-acquired pneumonia, or ARDS (577). Patients with risk factors for severe disease who present with a nonsevere influenza-like illness

**TABLE 8** Clinical features of patients with laboratory-confirmed A/2009/H1N1 infection

| Clinical symptom   | No. with symptom/total no. (% with symptom) |                                           |                                            |                                                   |                                                   |
|--------------------|---------------------------------------------|-------------------------------------------|--------------------------------------------|---------------------------------------------------|---------------------------------------------------|
|                    | Asia <sup>a</sup><br>( <i>n</i> = 2,077)    | Oceania <sup>b</sup><br>( <i>n</i> = 221) | Europe <sup>c</sup><br>( <i>n</i> = 1,943) | North America <sup>d</sup><br>( <i>n</i> = 3,857) | South America <sup>e</sup><br>( <i>n</i> = 3,236) |
| Fever              | 1,660/2,077 (79.9)                          | 187/221 (84.6)                            | 1,567/1,943 (80.6)                         | 3,340/3,841 (87.0)                                | 3,032/3,236 (93.7)                                |
| Cough              | 1,473/2,077 (70.9)                          | 153/167 (91.6)                            | 1,397/1,878 (74.4)                         | 2,274/2,701 (84.2)                                | 2,908/3,236 (89.9)                                |
| Sore throat        | 836/1,995 (41.9)                            | 30/55 (54.5)                              | 903/1,931 (46.8)                           | 1,379/3,408 (40.5)                                | 2,215/3,073 (72.1)                                |
| Running nose       | 577/1,938 (29.8)                            | 30/55 (54.5)                              | 518/882 (58.7)                             | 675/2,152 (31.4)                                  | 2,407/3,236 (74.4)                                |
| Headache           | 273/1,511 (18.1)                            | 4/12 (33.3)                               | 705/1,753 (40.2)                           | 674/2,203 (30.6)                                  | 1,015/1,465 (69.3)                                |
| Dyspnea            | 138/596 (23.2)                              | 16/43 (37.2)                              | 434/1,428 (30.4)                           | 1,444/2,371 (60.9)                                | 192/233 (82.4)                                    |
| Fatigue or malaise | 129/603 (21.4)                              | 23/43 (53.5)                              | 214/273 (78.4)                             | 112/159 (70.4)                                    | 1,370/1,782 (76.9)                                |
| Nausea or vomiting | 89/936 (9.5)                                | NM <sup>f</sup>                           | 371/1,700 (21.8)                           | 888/2,017 (44.0)                                  | 708/3,015 (23.5)                                  |
| Diarrhea           | 75/1,742 (4.3)                              | 30/209 (14.4)                             | 231/1,943 (11.9)                           | 716/3,016 (23.7)                                  | 318/2,138 (14.9)                                  |
| Myalgia            | 201/1,450 (13.9)                            | 58/109 (53.2)                             | 552/1,172 (47.1)                           | 867/2,118 (40.9)                                  | 827/1,331 (62.1)                                  |
| Arthralgia         | 10/271 (3.7)                                | NM                                        | 353/1,058 (33.4)                           | 20/44 (45.5)                                      | 10/11 (90.9)                                      |

<sup>a</sup> Including hospitalized patients (1,099 cases), critically ill patients (90 cases), clustering of an outbreak (11 cases), and other (877 cases) (6, 33, 58, 87, 111, 131, 231, 257, 317, 339, 350, 396, 420, 463, 469, 509, 511).

<sup>b</sup> Including hospitalized patients (166 cases), hospitalized pregnant patients (43 cases), and cystic fibrosis patients (12 cases) (142, 147, 183, 255).

<sup>c</sup> Including hospitalized patients (631 cases), clustering of 4 outbreaks in a camp and schools (172 cases), and other (1,140 cases) (53, 154, 223, 226, 246–248, 412, 480).

<sup>d</sup> Including hospitalized patients (1,782 cases), transplant recipients (242 cases), hospitalized pregnant patients (128 cases), clustering of 5 outbreaks in a plane, a camp, and schools (409 cases), and other (1,296 cases) (32, 65, 69, 74, 78, 130, 281, 282, 322, 347, 363, 365, 418, 514, 547, 550).

<sup>e</sup> Including hospitalized patients (204 cases), critically ill patients (29 cases), and other (3,003 cases) (199, 204, 355, 399, 442, 506, 578).

<sup>f</sup> NM, not mentioned.

may be managed as outpatients while waiting for the results of laboratory investigations. A chest radiograph should be performed if pulmonary involvement is suspected. The decision for hospitalization is based on the clinical assessment of disease severity, whether the patient can be readily followed up, and the risk factors for severe influenza infection. Patients with risk factors for severe disease should be empirically treated with oseltamivir or zanamivir, and patients with severe pneumonia should be treated with oseltamivir and broad-spectrum antibacterials for typical and atypical bacterial pneumonia (e.g., a  $\beta$ -lactam plus a macrolide). Influenza-like illness is clinically defined by the abrupt onset of fever and respiratory symptoms, such as rhinorrhea, sore throat, and cough, often with myalgia or headache; therefore, the specificity of this clinical definition can be as low as 10% during noninfluenza seasons and as high as 80% during an outbreak period. Additionally, the syndrome can be caused by other respiratory viruses, such as parainfluenza virus, enterovirus, adenovirus, and metapneumovirus, and several atypical bacteria, which can be differentiated only by laboratory testing.

### Specimen Collection

The best specimen for a laboratory diagnosis should contain a high viral titer and a large number of infected cells. The yield is

best when the specimens are collected within the first 2 to 3 days after the onset of symptoms during the peak of viral shedding in the respiratory tract (509). Aerosol-generating procedures for specimen collection should be performed using proper infection control measures with droplet precautions and eye shielding. The first specimen should be collected before the commencement of antiviral therapy whenever possible. For upper respiratory tract specimens, a nasopharyngeal aspirate, nasopharyngeal flocked swab, nasopharyngeal rayon swab, nasal wash fluid, throat wash fluid, or throat swab can be collected, in descending order of sensitivity (409). Lower respiratory tract specimens, including sputum, endotracheal aspirate, and bronchoalveolar lavage fluid, may be more sensitive in some cases with predominantly lower respiratory tract involvement (35). Additionally, influenza virus may be detected in blood, stool, or urine (509, 510, 516). Besides respiratory specimens, fecal specimens should also be collected in a viral transport medium, such as Hanks' balanced salt solution supplemented with 0.5% bovine serum albumin or 0.1% gelatin and antibiotics. Although the viral titer decreases with storage, the virus can usually be recovered by cell culture if it is maintained in a viral transport medium at 4°C for up to 5 days. A longer

**TABLE 9** Complications in hospitalized patients with laboratory-confirmed A/2009/H1N1 infection

| Outcome                | No. with outcome/total no. (% with outcome) <sup>a</sup> |                           |                            |                                   |                                 |
|------------------------|----------------------------------------------------------|---------------------------|----------------------------|-----------------------------------|---------------------------------|
|                        | Asia ( <i>n</i> = 1,848)                                 | Oceania ( <i>n</i> = 617) | Europe ( <i>n</i> = 2,386) | North America ( <i>n</i> = 2,097) | South America ( <i>n</i> = 251) |
| Admission to ICU       | 126 (6.8)                                                | 95/617 (15.4)             | 235 (9.8)                  | 486/1,773 (27)                    | 47 (18.8)                       |
| Mechanical ventilation | 67 (3.6)                                                 | 25/222 (11.3)             | NM <sup>b</sup>            | 308/1,773 (17.4)                  | 42 (16.7)                       |
| Death                  | 30 (1.6)                                                 | 14/617 (2.3)              | 57 (2.4)                   | 145/2,097 (6.9)                   | 13 (5.2)                        |

<sup>a</sup> The chi-square test was used to assess the difference in outcomes (admission to ICU, mechanical ventilation, and death) among hospitalized patients. There were significantly higher rates of admissions to ICU ( $P < 0.001$ ), mechanical ventilation ( $P < 0.001$ ), and death ( $P < 0.001$ ) in North America than in Asia. References are as follows: for patients in Asia, references 131, 493, and 511; for patients in Oceania, references 142, 147, 177, 237, and 527; for patients in Europe, references 132 and 523; for patients in North America, references 32, 69, 75, 281, 322, 365, and 418; and for patients in South America, reference 355.

<sup>b</sup> NM, not mentioned.

**TABLE 10** Extrapulmonary manifestations in patients with laboratory-confirmed A/2009/H1N1 infection

| System involved (illness)                                 | Description <sup>a</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Reference |
|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Liver (hepatitis)                                         | F/37 yr and M/57 yr with hypertension, had elevated ALT of 1.5–2 times between days 7 and 10 after ICU admission; postmortem liver biopsy of the first patient showed micro- and macrovesicular steatosis, and RT-PCR was positive for influenza A H1N1 virus                                                                                                                                                                                                                                                                                                                                     | 61        |
| Heart (myocarditis)                                       | Four H1N1 influenza virus-associated myocarditis cases based on elevated cardiac enzymes ( $n = 2$ ), significant acute decrease in left ventricular systolic function demonstrated by the echocardiogram ( $n = 3$ ), or histologic evidence of severe myocarditis ( $n = 1$ ); three children presented with fulminant myocarditis, 1 with a fatal outcome and 2 requiring ECMO support                                                                                                                                                                                                         | 42        |
|                                                           | Seven patients with H1N1 influenza virus-associated myocarditis, aged 3–52 yr, with 2 fatal outcomes and 3 requiring ECMO, one requiring MCS, and one requiring IABP                                                                                                                                                                                                                                                                                                                                                                                                                              | 323       |
|                                                           | 6.8% (5/74) of patients had myocarditis, and all died                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 510       |
| Heart (reversible cardiac dysfunction)                    | 4.9% (6/123) of patients had either new or worsened left ventricular dysfunction; age ranged from 23 to 51 yr; all had preexisting medical conditions; ICU care was required for 83% (5/6); 67% (4/6) improved on follow-up echocardiograms                                                                                                                                                                                                                                                                                                                                                       | 376       |
| Central nervous system (acute necrotizing encephalopathy) | F/3-yr-old Italian patient had high fever and severe convulsions following 2 days of cough and diarrhea; lumbar puncture showed slightly increased pressure (170 mm H <sub>2</sub> O); CSF contained 6 cells/mm <sup>3</sup> and elevated protein, numerous red cells, and normal lactate and pyruvate levels; MRI demonstrated abnormal signal intensity in the supratentorial white matter, splenium of the corpus callosum, dorsal aspect of pons, bilateral thalami, and cerebellar hemispheres with mass effect on the surrounding parenchyma suggestive of acute necrotizing encephalopathy | 419       |
| Central nervous system (encephalopathy)                   | Two unrelated Chinese patients, aged 3 and 4 yr, had thermolabile CPT-II variants that were associated with persistent high-fever-triggered viral infection-associated encephalopathy, multiorgan failure, and death                                                                                                                                                                                                                                                                                                                                                                              | 370       |
| Muscle                                                    | M/56 yr receiving therapy for recurrent multiple myeloma had rhabdomyolysis with myoglobinuria that arose during convalescence from severe A/2009/H1N1 pneumonia                                                                                                                                                                                                                                                                                                                                                                                                                                  | 224       |
| Kidney                                                    | 50 critically ill patients with severe respiratory syndrome (47 confirmed cases, 3 probable cases); kidney injury, kidney failure, and need for dialysis occurred in 66.7%, 66%, and 11% of patients, respectively; kidney failure was associated with increased death (OR, 11.29; 95% CI, 1.29–98.9), whereas the need for dialysis was associated with an increase in length of stay (RR, 2.38; 95% CI, 2.13–25.75)                                                                                                                                                                             | 485       |
| Vessel (vascular thrombosis)                              | 7 (5.9%) of patients (18–84 yr), experienced thrombotic vascular event, involving coronary and infrarenal aorta, and femoral and iliac veins with pulmonary embolism, between 3 and 33 days after symptom onset                                                                                                                                                                                                                                                                                                                                                                                   | 49        |
|                                                           | 1.4% (1/74) of patients had thrombosis in branches of pulmonary artery                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 510       |
| Reticuloendothelial system (reactive hemophagocytosis)    | 2.7% (2/74) of patients developed reactive hemophagocytosis in lymph node or bone marrow                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 510       |

<sup>a</sup> ALT, alanine aminotransferase; CPT-II, carnitine palmitoyltransferase II; CSF, cerebrospinal fluid; ECMO, extracorporeal membrane oxygenation; MRI, magnetic resonance imaging; F, female; IABP, intra-aortic balloon pump; ICU, intensive care unit; M, male; MCS, mechanical circulatory support; OR, odds ratio; CI, confidence interval; RR, relative risk.

period of storage requires a temperature of  $-70^{\circ}\text{C}$  to maintain viral viability (21).

### Nucleic Acid Amplification

RT-PCR diagnostic tests are often targeted against the M gene, which is abundant and genetically conserved, and the H gene, which is abundant and subtype specific (153, 331, 447). The most useful laboratory test for the clinical management of A/2009/H1N1 influenza is nucleic acid amplification after reverse transcription (Table 11). In 10 studies, the overall reported sensitivity of real-time RT-PCR ranged from 90.5% to 97.6% (60, 202, 240, 286, 321, 336, 404, 422, 432, 559). The majority of the studies reported a specificity of 100%, with four exceptions (60, 404, 432, 559). Many diagnostic laboratories and commercial vendors have rapidly responded to this pandemic by providing in-house or commercially packed real-time nucleic acid amplification assays.

### Antigen Detection

All commercially available immunochromatographic assays for the detection of the viral NP can differentiate between influenza A

and B viruses but not between the different subtypes. The reported sensitivities ranged from 11% to 83.7% with a median of 53.3%, which is significantly inferior to that of RT-PCR or viral culture. The specificities for immunochromatographic assays ranged from 96% to 100% (Table 12) (115–117, 144, 157, 171, 191, 201, 242, 309, 315, 497, 525, 526, 540). Similarly, the direct immunofluorescent antigen detection test for the influenza virus NP had a sensitivity ranging from 38.7% to 92.8%, with a median of 64.9%, and a specificity of 94.5% to 100% (191, 202, 242, 446). However, the testing of these exfoliated cells must be performed soon after specimen collection. The cells from specimens can be washed in cold buffer and treated with *N*-acetylcysteine to remove mucus before fixation onto slides (82, 83). In some laboratories, the influenza A and B viruses, respiratory syncytial virus (RSV), adenovirus, and parainfluenza (PIF) 1, 2, and 3 viruses are concurrently detected in a common pool before a monoclonal antibody is used to differentiate the respiratory virus that is present (82, 83).

### Viral Culture

The gold standard for the diagnosis of influenza virus relies on a positive culture of the respiratory secretions in cell lines, such as

TABLE 11 Evaluation and clinical utilization of molecular diagnostic tests for A/2009/H1N1 virus<sup>a</sup>

| Test mode (commercial kit), target                                                                                                   | Samples (clinical, RNA extract, or virus isolates), patient demographic (if available), and country | Analytical performance                                                                                                                                                                                                                                                                                    | Reference |
|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Real-time RT-PCR (Luminex xTAG respiratory virus panel; Luminex Molecular Diagnostics, Toronto, Canada)                              | 288 nasopharyngeal specimens (flocked swabs), aged 4 days-to 98 yr, USA                             | Sensitivity, 97.8%; specificity, 100%; PPV, 100%; NPV, 97.3%; gold standard, real-time RT-PCR by CDC protocol                                                                                                                                                                                             | 202       |
| In-house real-time RT-PCR, H1 gene                                                                                                   | 38 clinical samples, <sup>b</sup> Canada                                                            | Detection limit, virus in a 10 <sup>6</sup> dilution of 4 × 10 <sup>6</sup> TCID <sub>50</sub> /ml when 5 µl was used as the template; median (range) of C <sub>T</sub> values, 25.9 (17–45); sensitivity, 95.2%; specificity, 100%; gold standard, amplification and direct sequencing of the products   | 422       |
| In-house real-time RT-PCR, M1 gene                                                                                                   | 36 clinical samples, <sup>b</sup> Canada                                                            | Detection limit, virus in a 10 <sup>6</sup> dilution of 4 × 10 <sup>6</sup> TCID <sub>50</sub> /ml when 5 µl was used as the template; median (range) of C <sub>T</sub> values, 29.1 (19.5–45); sensitivity, 90.5%; specificity, 100%; gold standard, amplification and direct sequencing of the products | 422       |
| In-house real-time RT-PCR, M2 gene                                                                                                   | 38 clinical samples, <sup>b</sup> Canada                                                            | Detection limit, virus in a 10 <sup>6</sup> dilution of 4 × 10 <sup>6</sup> TCID <sub>50</sub> /ml when 5 µl was used as the template; median (range) of C <sub>T</sub> value: 27.8 (18.8–45); sensitivity, 97.6%; specificity, 100%; gold standard, amplification and direct sequencing of the products  | 422       |
| In-house duplex RT-PCR, annealing temp at 55°C                                                                                       | 198 consecutive nasopharyngeal, nasal, and throat swabs, Canada                                     | Sensitivity, 93.9%; specificity, 100%; PPV, 100%; NPV, 97.1%; a modified “gold standard” was used <sup>c</sup>                                                                                                                                                                                            | 336       |
| In-house duplex RT-PCR, annealing temp at 50°C                                                                                       | 198 consecutive nasopharyngeal, nasal, and throat swabs, Canada                                     | Sensitivity, 77.3%; specificity, 100%; PPV, 100%; NPV, 89.8%; a modified “gold standard” was used <sup>c</sup>                                                                                                                                                                                            | 336       |
| In-house monoplex RT-PCR                                                                                                             | 198 consecutive nasopharyngeal, nasal, and throat swabs, Canada                                     | Sensitivity, 80.3%; specificity, 100%; PPV, 100%; NPV, 91.0%; a modified “gold standard” was used <sup>c</sup>                                                                                                                                                                                            | 336       |
| In-house real-time RT-PCR                                                                                                            | 198 consecutive nasopharyngeal, nasal, and throat swabs, Canada                                     | Sensitivity, 90.9%; specificity, 100%; PPV, 100%; NPV, 95.7%; a modified “gold standard” was used <sup>c</sup>                                                                                                                                                                                            | 336       |
| In-house RT-LAMP, H1 gene                                                                                                            | 260 clinical samples <sup>b</sup> collected from Japanese and Vietnamese patients                   | Detection limit, 10 RNA copies per reaction volume; sensitivity, 97.8%; specificity, 100%; gold standard, real-time RT-PCR                                                                                                                                                                                | 321       |
| RT-LAMP (Eiken Chemical, Tokyo, Japan), H1 gene and M gene                                                                           | 45 nasal swabs, Japan                                                                               | Sensitivity, 96.3%; specificity, 88.9%; gold standard, real-time RT-PCR                                                                                                                                                                                                                                   | 404       |
| RT-LAMP (Eiken Chemical, Tokyo, Japan), H1 gene                                                                                      | 56 nasal swabs, mean age of 31.6 yr (20–51 yr), Japan                                               | Detection limit, 100 copies of virus; sensitivity, 96.3%; gold standard, real-time RT-PCR                                                                                                                                                                                                                 | 240       |
| StepOnePlus real-time RT-PCR (Applied Biosystems, Foster City, CA), H1 gene                                                          | 50 respiratory specimens (nasopharyngeal aspirates, nasopharyngeal swabs, throat swabs), Hong Kong  | Detection limit, 1,252 gene copy equivalent                                                                                                                                                                                                                                                               | 328       |
| In-house multiplex PCR, H1 and H3 genes                                                                                              | 50 respiratory specimens (nasopharyngeal aspirates, nasopharyngeal swabs, throat swabs), Hong Kong  | Detection limit, 125.2 gene copy equivalent                                                                                                                                                                                                                                                               | 328       |
| In-house conventional RT-PCR, H1 gene                                                                                                | 50 respiratory specimens (nasopharyngeal aspirates, nasopharyngeal swabs, throat swabs), Hong Kong  | Detection limit, 12.52 gene copy equivalent                                                                                                                                                                                                                                                               | 328       |
| In-house nucleic acid dipstick test with isothermal amplification and visual detection on dipstick, H1 gene                          | 262 nasal or throat swabs; mean age, 22.4 yr (19 days–91 yr), UK                                    | Sensitivity, 95.3%; specificity, 99.4%; PPV, 98.8%; NPV, 97.8%; gold standard, real-time RT-PCR                                                                                                                                                                                                           | 559       |
| Nucleic acid lateral-flow assay based on rapid amplification and hybridization technology (Analytik Jena AG, Jena, Germany), H1 gene | 174 nasal swabs, Japan                                                                              | Sensitivity, 88%; specificity, 94%; PPV, 96%; NPV, 84%; gold standard: real-time RT-PCR                                                                                                                                                                                                                   | 432       |
| In-house real-time RT-PCR, segment 7 carrying the M1 gene                                                                            | 11 clinical samples, <sup>b</sup> Ireland                                                           | Detection limit, detect virus in 10 <sup>-3</sup> dilution of a clinical sample; sensitivity, 100%; specificity, 88.8%; gold standard, confirmation by reference laboratory in Health Protection Agency                                                                                                   | 60        |
| In-house conventional 1-step RT-PCR, H1 gene                                                                                         | Test evaluation, Hong Kong                                                                          | Detection limits for the positive control in the range of 1.0 × 10 <sup>-4</sup> of the TCID <sub>50</sub> per reaction; specificity, 100%                                                                                                                                                                | 447       |
| In-house real-time RT-PCR, H1 gene                                                                                                   | Test evaluation, Hong Kong                                                                          | Detection limits for the positive control in the range of 2.0 × 10 <sup>-3</sup> of the TCID <sub>50</sub> per reaction; specificity, 100%                                                                                                                                                                | 447       |
| In-house real-time RT-PCR, H1 gene                                                                                                   | Test evaluation, Taiwan                                                                             | Detection limit, 10 copies of H gene per reaction                                                                                                                                                                                                                                                         | 568       |
| In-house real-time RT-PCR, H1 gene                                                                                                   | Test evaluation, Germany                                                                            | Detection limit, 100–1,000 genomic copies per ml                                                                                                                                                                                                                                                          | 542       |
| In-house real-time RT-PCR, N gene                                                                                                    | Test evaluation, Germany                                                                            | Detection limit, 100–1,000 genomic copies per ml                                                                                                                                                                                                                                                          | 542       |
| In-house real-time RT-PCR, modified primer to detect mutation H275Y                                                                  | Test evaluation, Israel                                                                             | Analytical sensitivity, 0.014 TCID <sub>50</sub> with 97.2% amplification efficiency                                                                                                                                                                                                                      | 260       |
| Commercial POC molecular test (Xpert Flu A Panel nucleic acid amplification-based POC test; Cepheid, Sunnyvale, CA)                  | Test evaluation of 7 clinical samples <sup>b</sup> positive for influenza virus, the Netherlands    | Sensitivity, 71.4%; specificity, 100%; gold standard: real-time RT-PCR                                                                                                                                                                                                                                    | 286       |

<sup>a</sup> H, hemagglutinin; N, neuraminidase; M, matrix; NP, nucleoprotein; PPV, positive predictive value; NPV, negative predictive value; POC, point of care; RT-LAMP, reverse transcription–loop-mediated isothermal amplification; RT-PCR, reverse transcription-PCR; TCID<sub>50</sub>, 50% tissue culture infective dose; C<sub>T</sub>, threshold cycle.

<sup>b</sup> The nature of the clinical samples was not specified.

<sup>c</sup> A modified “gold standard” was used to assess the clinical performance of all RT-PCR assays when a positive case was defined by concordant results between at least two RT-PCRs targeting different genomic regions and subsequent sequence analysis to ensure the specificities of the primers.



TABLE 12 Evaluation and clinical utilization of conventional diagnostic tests for A/2009/H1N1 virus<sup>a</sup>

| Test mode (commercial kit), target                                             | Samples (clinical, RNA extract, or virus isolates), patient demographic (if available), and country                | Analytical performance                                                                                                          | Reference |
|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-----------|
| Rapid antigen detection (Binax Now A+B)                                        | 144 clinical samples, <sup>b</sup> median age of 18 yr (1–59 yr), Germany                                          | Sensitivity, 11.1%; gold standard, real-time RT-PCR for H gene                                                                  | 157       |
|                                                                                | 84 nasopharyngeal specimens, USA                                                                                   | Sensitivity, 38.3%; specificity, 100%; PPV, 100%; NPV, 88.2%; gold standard, real-time RT-PCR using Luminex xTAG RVP (Luminex)  | 525       |
|                                                                                | 820 nasopharyngeal specimens (flocked swabs), median age of 3.4 years (1 mo–17 yr), Canada                         | Sensitivity, 62%; specificity, 99%; gold standard, conventional RT-PCR                                                          | 242       |
|                                                                                | 354 nasopharyngeal specimens (flocked swabs), median age of 43 yr (18–89 yr), Spain                                | Sensitivity, 32%; specificity, 100%; PPV, 100%; NPV, 67%; gold standard, real time RT-PCR (Applied Biosystems real-time RT-PCR) | 201       |
|                                                                                | 254 nasopharyngeal specimens (flocked swabs), median age of 14.1 yr (7 mo–53 yr), South Korea                      | Sensitivity, 83.7%; specificity, 100%; gold standard: real-time RT-PCR                                                          | 117       |
|                                                                                | 1 nasopharyngeal aspirate specimen collected on day 3 of illness, Hong Kong                                        | Detection limit, 6.8 copies M gene/ml                                                                                           | 81        |
| Rapid antigen detection (Directigen EZ A+B)                                    | 84 nasopharyngeal specimens, USA                                                                                   | Sensitivity, 46.7%; specificity, 100%; PPV, 100%; NPV, 89.6%; gold standard, real-time RT-PCR using Luminex xTAG RVP (Luminex)  | 525       |
|                                                                                | 1 nasopharyngeal aspirate specimen collected on day 3 of illness, Hong Kong                                        | Detection limit, 6.1 copies M gene/ml                                                                                           | 81        |
| Rapid antigen detection (Espline)                                              | 1 nasopharyngeal aspirate specimen collected on day 3 of illness, Hong Kong                                        | Detection limit, 5.8 copies M gene/ml                                                                                           | 81        |
| Rapid antigen detection (QuickVue A+B)                                         | 39 clinical samples, <sup>b</sup> USA                                                                              | Sensitivity, 51%; specificity, 99%; gold standard, real-time RT-PCR                                                             | 171       |
|                                                                                | 84 nasopharyngeal specimens, USA                                                                                   | Sensitivity, 53.3%; specificity, 100%; PPV, 100%; NPV, 90.8%; gold standard, real-time RT-PCR using Luminex xTAG RVP (Luminex)  | 525       |
|                                                                                | 174 nasal or throat swabs, Australia                                                                               | Sensitivity, 53.4%; specificity, 100%; PPV, 100%; NPV, 76.2%; gold standard, real-time in-house RT-PCR using a TaqMan probe     | 315       |
|                                                                                | 418 patients (aged 6 mo–14 yr) with nasal and throat swabs, Thailand                                               | Sensitivity, 62.7%; specificity, 99.2%; gold standard, real-time RT-PCR                                                         | 497       |
|                                                                                | 360 nasal specimens, median age of 13.7 yr (6 mo–73 yr), Philippines                                               | Sensitivity, 63%; specificity, 96%; PPV, 97%; NPV, 57%; gold standard, conventional RT-PCR                                      | 526       |
|                                                                                | 526 respiratory specimens (nasopharyngeal swabs, pharyngeal washes, bronchoalveolar lavage fluid samples), Germany | Sensitivity, 18.2%; specificity, 100%; PPV, 100%; NPV, 78.1%; gold standard, real-time RT-PCR                                   | 191       |
|                                                                                | 1 nasopharyngeal aspirate specimen collected on day 3 of illness, Hong Kong                                        | Detection limit, 6.5 copies M gene/ml                                                                                           | 81        |
| Rapid antigen detection (Binax Now A+B and 3 M Rapid Detection Flu A+B test)   | 288 nasopharyngeal specimens (flocked swabs), aged 4 days–98 yr, USA                                               | Sensitivity, 17.8%; specificity, 93.6%; PPV, 77.4%; NPV, 47.9%; gold standard, real-time RT-PCR                                 | 202       |
| Rapid antigen detection <sup>c</sup>                                           | 1,599 clinical samples <sup>b</sup> from pediatric patients, USA                                                   | Sensitivity, 17.8%; specificity, 93.6%; gold standard, real-time RT-PCR using Luminex xTAG RVP (Luminex)                        | 540       |
| Rapid antigen detection (SD Bioline Influenza Ag Standard Diagnostics for H)   | 759 nasopharyngeal specimens (flocked swabs), aged 2 wk–83 yr, South Korea                                         | Sensitivity, 77%; specificity, 100%; PPV, 100%; NPV, 86%; gold standard, real-time RT-PCR                                       | 116       |
| Rapid antigen detection [SD Bioline Influenza Ag A/B/(H1N1) Pandemic for H]    | 260 clinical samples, <sup>b</sup> aged 2 mo–78 yr, South Korea                                                    | Sensitivity, 70%; specificity, 98.4%; PPV, 94.3%; NPV, 89.7%; gold standard, real-time RT-PCR                                   | 309       |
| Rapid antigen detection (SD Bioline Influenza Ag, Standard Diagnostics for NP) | 260 clinical samples, <sup>b</sup> aged 2 mo–78 yr, South Korea                                                    | Sensitivity, 58.8%; specificity, 99.6%; PPV, 98.1%; NPV, 86.5%; gold standard, real-time RT-PCR                                 | 309       |
| Rapid antigen detection (SD Bioline Influenza Ag)                              | 254 nasopharyngeal specimens (flocked swabs), median age of 14.1 yr (7 mo–53 yr), South Korea                      | Sensitivity, 69.5%; specificity, 100%; gold standard, real-time RT-PCR                                                          | 117       |
|                                                                                | 938 throat or nasopharyngeal swabs, all patient over 15 yr, South Korea                                            | Sensitivity, 44%; specificity, 99.9%; gold standard, real-time RT-PCR                                                           | 115       |
| Rapid antigen detection (Wondfo)                                               | 1 nasopharyngeal aspirate specimen collected on day 3 of illness, Hong Kong                                        | Detection limit 5.8 copies M gene/ml                                                                                            | 81        |
| Rapid antigen detection (ClearView Exact Influenza A & B)                      | 1,016 oropharyngeal and nasopharyngeal swabs, median age of 45 yr (3 mo–97 yr), Spain                              | Sensitivity, 19%; specificity, 100%; PPV, 100%; NPV, 75%; gold standard, real time RT-PCR                                       | 144       |
| Rapid antigen detection (DFA test)                                             | 288 nasopharyngeal specimens (flocked swabs), aged 4 days–98 yr, USA                                               | Sensitivity, 46.7%; specificity, 94.5%; PPV, 91.3%; NPV, 58.9%; gold standard, real time RT-PCR                                 | 202       |
|                                                                                | 111 nasopharyngeal specimens (109 flocked swabs and 3 aspirates), median age of 44.1 yr, USA                       | Sensitivity, 92.8%; specificity, 97.1%; gold standard, real-time RT-PCR                                                         | 446       |
|                                                                                | 820 nasopharyngeal specimens (flocked swabs), median age of 3.4 yr (1 mo–17 yr), Canada                            | Sensitivity, 83%; specificity, 96%; gold standard, conventional RT-PCR                                                          | 242       |
|                                                                                | 526 respiratory specimens (nasopharyngeal swabs, pharyngeal washes, bronchoalveolar lavage fluid samples), Germany | Sensitivity, 38.7%; specificity, 100%; PPV, 100%; NPV, 82.2%; gold standard, real-time RT-PCR                                   | 191       |
| ELISA <sup>d</sup> for IgG antibodies                                          | 783 serum samples, India                                                                                           | Compared to hemagglutination inhibition test, concordance of 98.4%                                                              | 14        |
| Immunochromatography, monoclonal antibodies against NP                         | Clinical samples <sup>b</sup> from 5 patients with PCR-confirmed pandemic influenza A (H1N1), Japan                | Detection limit, $2 \times 10^5$ copies/kit; sensitivity, 100%; specificity, 100%; gold standard, real time RT-PCR              | 389       |
| Viral culture (R-mix viral culture)                                            | 288 nasopharyngeal specimens (flocked swabs), aged 4 days–98 yr, USA                                               | Sensitivity, 88.9%; specificity, 100%; PPV, 100%; NPV, 87.9%; gold standard, real-time RT-PCR                                   | 202       |
| Viral culture, MDCK cells                                                      | 526 respiratory specimens (nasopharyngeal swabs, pharyngeal washes, bronchoalveolar lavage fluid samples), Germany | Sensitivity, 45.7%; specificity, 99.8%; PPV, 95.5%; NPV, 94.8%; gold standard, real-time RT-PCR                                 | 191       |

<sup>a</sup> DFA, direct fluorescent antibody; H, hemagglutinin; MDCK, Madin-Darby canine kidney; M, matrix; N, neuraminidase; NP, nucleoprotein; PPV, positive predictive value; NPV, negative predictive value; RT-PCR, reverse transcription-PCR; ELISA, enzyme-linked immunosorbent assay.

<sup>b</sup> The nature of the clinical samples was not specified.

<sup>c</sup> The details of rapid antigen detection were not specified.

<sup>d</sup> Recombinant hemagglutinin protein-based enzyme-linked immunosorbent assay.

the Madin-Darby canine kidney (MDCK) or rhesus monkey kidney cell line, in the presence of trypsin (329). Most continuous cell lines do not produce proteases that cleave H to make infectious viral progeny, with the exception of Caco2 cells, which can produce trypsin-like proteases (349). Tosylsulfonyl phenylalanyl chloromethyl ketone (TPCK)-treated trypsin at 1 to 2  $\mu\text{g}/\text{ml}$  is added to the cell culture medium. TPCK treatment will inactivate the chymotrypsin in a pancreatic extract, whose proteolytic activity on H will nullify the trypsin-mediated enhancement of viral infectivity. Fetal calf sera contain inhibitory factors that must be removed by washing with Hanks' buffer prior to infection. Incubation with serum-free culture medium with trypsin at 33 to 35°C will usually produce a cytopathic effect within 1 to 7 days. Additional strains may be recovered with prolonged incubation of up to 14 days after one blind passage. The interval between inoculation and detection ranges from 2 to 14 days, with a median of 3 to 5 days (346). The R-mix cell line (a mixture of mink lung and A549 cell lines) was reported to have a sensitivity of 88.9%, whereas the MDCK cell line had a sensitivity of only 45.7%, compared with real-time RT-PCR as the gold standard (191). In addition to identifying the microscopic changes of cytopathic effects, positive cultures can be detected by hemadsorption. Influenza virus isolates can be typed as type A or B and subtyped by immunofluorescence. In contrast to nucleic acid amplification, antigen detection, and antibody assays, viral culture is less affected by genetic or antigenic changes. Cell culture had a specificity of nearly 100% (191, 202).

### Antibody Testing

The detection of specific antibodies in serum may be used when the specimens for viral isolation or RNA or antigen detection are negative, inadequate, or unavailable. Antibody testing has detected up to one-third of severe cases admitted to ICUs when the RT-PCR results were negative (277). However, this testing cannot differentiate different lineages of the same subtype of the influenza A virus. The majority of individuals have already been exposed to the seasonal influenza A virus, which has almost identical NP and matrix protein sequences; therefore, a 4-fold rise in the complement fixation antibody titer with these antigens indicates a recent influenza A virus infection without differentiating between seasonal H1N1 or H3N2 infection and the pandemic 2009 H1N1 infection (14). Although recombinant H has been found to be sensitive and specific for antibody detection by EIA, cross-reactivity between seasonal H1N1 and pandemic H1N1 may still occur, because their overall amino acid identity was 80.1% for H (440 out of 549 amino acids), 72% for HA1, and 92% for HA2 (Fig. 1). Therefore, IgG and IgM detection by EIA is generally not useful, except in the case of avian H5N1 infection in humans, where recombinant H5 was used in a Western blot immunoassay (44). The traditional gold standard techniques for detecting influenza virus-specific antibodies are neutralization and hemagglutination inhibition assays, which may detect subtype-specific or lineage-specific serological responses. Human influenza viruses can agglutinate turkey, human, and guinea pig erythrocytes; however, most laboratories routinely use turkey erythrocytes for hemagglutination inhibition antibody tests. Turkey erythrocytes are small and nucleated and sediment quickly, producing a clear and reproducible endpoint. Turkey erythrocytes express a mixture of  $\alpha$ -2,3- and  $\alpha$ -2,6-linked sialic acid receptors and can be used for the detection of antibodies against either human or avian influenza viruses, whereas the avian influenza A H5N1, H9N2 and H7N7

viruses agglutinate horse erythrocytes, which almost exclusively express  $\alpha$ -2,3-linked sialic acid receptors (273). Antibody levels accurately correlate with protection from (or susceptibility to) disease and vaccination status (84). A definitive serological diagnosis of acute influenza requires the demonstration of a 4-fold increase in antibody titers on paired acute- and convalescent-phase serum samples. A seroprevalence study using the hemagglutination inhibition test was useful in 2009 when the population was immunologically naïve except for elderly patients, who may have been previously exposed to the 1976 swine influenza virus or the 1918 Spanish influenza virus, both of which have cross-reactivity to the A/2009/H1N1 virus (45). Since changes in the H proteins of influenza A H3N2 and influenza B viruses have markedly diminished their ability to agglutinate avian erythrocytes, including turkey erythrocytes since late 1990s, the neutralizing antibody test may be a better option for these viruses.

Though it is more labor-intensive and still could be confounded by cross-reactivity, the neutralizing antibody test is often considered the gold standard serological test for immunity against different strains of influenza virus. A retrospective analysis of the serum antibody responses assayed by neutralizing antibody titers against the A/2009/H1N1 virus in 881 convalescent patients demonstrated that 90% of them had a seroprotective titer of 1:40 or above (265). A multivariate analysis by ordinal regression showed that pneumonia and sputum production were two independent factors associated with higher levels of convalescent-phase neutralizing antibody titers. Patients who were afebrile on presentation were associated with a subsequent poor neutralizing antibody titer ( $<1:40$ ). A positive correlation between the nasopharyngeal viral load on presentation and the convalescent neutralizing antibody titer was demonstrated. Convalescent patients with a high neutralizing antibody titer were suitable candidates for the donation of plasma for passive immunotherapy (265).

### Antiviral Susceptibility Testing

Viral susceptibility to adamantanes can be detected by standard cell protection assays, whereas the inhibition of the enzymatic activity of N is the most sensitive and specific phenotypic means of detecting resistance to neuraminidase inhibitors, such as oseltamivir, peramivir, and zanamivir. The standard methods for detecting antiviral resistance based on changes in the viral phenotype in cell culture, such as the plaque reduction and yield reduction assays, were not reliable for the detection of clinical isolates resistant to neuraminidase inhibitors (217, 218). The contradictory findings from cell culture- and enzymatic activity-based assays suggested that the H of clinical isolates may bind suboptimally to the  $\alpha$ -2,3-linked sialic receptors of the MDCK cells used in cell protection assays, because human influenza viruses bind preferentially to  $\alpha$ -2,6-linked sialic acid. When H binds with a lower affinity to MDCK cell receptors, the virus is less dependent on the neuraminidase activity of N for its release; therefore, the virus appears to be less sensitive to neuraminidase inhibitors *in vitro*. Zanamivir susceptibility in animal challenge models correlated well with *in vitro* susceptibility as determined by the neuraminidase inhibition assay but not by the plaque reduction assay in MDCK cells (507). The majority of laboratories currently use a phenotypic assay with chemiluminescent or fluorescent substrates (NAStar; Applied Biosystems, Foster City, CA) to detect the susceptibility of the enzymatic activity of N to neuraminidase inhibitors.

A rapid genotypic assay for the detection of mutations causing resistance against adamantanes and neuraminidase inhibitors can be performed by standard PCR sequencing and, more recently, by pyrosequencing without the need for viral culture (324). Multiplex PCR can simultaneously detect the presence of the A/2009/H1N1 virus and the mutations responsible for the resistance (29, 368). Methods that can detect the resistant strains among a mixed population include pyrosequencing (146), high-resolution melting curve analysis (102), and, in research laboratories, deep sequencing (197). Other methods include RT-PCR and the restriction fragment length polymorphism assay (415). However, pyrosequencing can detect the emergence of resistant quasiespecies in treated individuals without laborious cloning and Sanger sequencing. The clinical significance of viral quasiespecies in drug resistance is undetermined except in immunosuppressed hosts in whom resistant viral quasiespecies have emerged early in treatment (22). In this group of patients, prolonged viral carriage and shedding provided an opportunity for the viruses to pick up compensatory mutations that maintain the resistance after selective drug pressure is removed.

## ANTIVIRAL AND IMMUNOMODULATORY THERAPIES

### Antiviral Treatment

The antivirals against influenza A virus that are currently available and commonly used include the adamantanes (amantadine and rimantadine) and neuraminidase inhibitors (oseltamivir, zanamivir, and peramivir), but only the latter are active against the A/2009/H1N1 virus. Oseltamivir is available orally. Zanamivir is available either as a dry powder that is delivered by oral inhalation or, more recently, as an intravenous formulation. Nebulized zanamivir is useful in selected cases (135). However, nebulization of the powder form of zanamivir that is dissolved in water has been found to clog the ventilator circuit and should not be used (179). Intravenous peramivir was available only for compassionate use in the United States during the pandemic (34). In addition to neuraminidase inhibitors, the traditional Chinese medicine maxingshigan-yinqiaosan has been shown to hasten fever resolution in a randomized trial, but its action may be immunomodulatory rather than antiviral (534). The role of other newer antivirals, such as nucleozin analogs (295), the neuraminidase inhibitor CS-8958 (311), the polymerase inhibitors viramidine (474) and T-705 (312), and sialidase DAS181 (27), requires further evaluation (243). Combination therapy with the protease inhibitor aprotinin and other antivirals may be effective, but the side effects of aprotinin treatment, including stroke, heart attack, kidney failure, thrombosis, and anaphylaxis, must be considered (582).

Randomized controlled trials with patients having seasonal influenza suggested that neuraminidase inhibitors shortened the duration of illness by approximately 1 day (245). However, most authorities would not recommend routine oseltamivir treatment for mild illness because of the risk of fostering drug resistance. For the A/2009/H1N1 virus, oseltamivir therapy has generally been associated with a faster resolution of symptoms and a more rapid clearance of viral shedding in hospitalized patients of all ages, organ transplant recipients, and pregnant women (350, 357, 575) (Table 13). A longer delay of oseltamivir treatment in severe cases was associated with a poorer outcome (281, 574) because viral shedding in the respiratory tract peaks at 24 to 72 h after infection.

Since viral shedding is already substantial even before the onset of symptoms, the efficacy of treatment decreases sharply with any delay in antiviral administration.

Intravenous zanamivir and peramivir have been used in patients with severe disease (Table 14). These antivirals have *in vitro* activity against the A/2009/H1N1 virus and have been beneficial in some cases; however, no randomized controlled trials or comparative clinical trials have been reported (238, 303). Oseltamivir-zanamivir combination therapy has not been studied in patients with A/2009/H1N1 influenza, but it has been found to be less effective than oseltamivir monotherapy for the treatment of seasonal influenza (161). The optimal dosing for intravenous zanamivir is not known, but 600 mg every 12 h has been shown to be effective for oseltamivir-resistant strains of the A/2009/H1N1 virus in an *in vitro* hollow-fiber infection model (47).

Ribavirin has been used for the treatment and prophylaxis of influenza (90). The routes of administration include oral, intravenous, and aerosolization routes. Anecdotal reports have demonstrated the efficacy of this therapy against influenza A and B, including infections in immunosuppressed hosts. Unfortunately, a consistent benefit has not been observed in all of the clinical trials, and intravenous ribavirin is currently not considered a drug of choice for the treatment of influenza because of its toxicity. Similar to the case for inhaled zanamivir, inhaled ribavirin may not be able to penetrate the poorly aerated consolidations of pneumonic lungs.

### Antiviral Resistance

All A/2009/H1N1 strains are resistant to adamantanes due to the S31N mutation in the M2 protein (79, 554). Globally, oseltamivir resistance due to the H275Y (by N1 numbering) substitution in neuraminidase accounts for fewer than 2% of the strains tested (554). In addition, H275Y strains are resistant to peramivir but remain susceptible to zanamivir. The differential antiviral susceptibility is due to the differences between the binding mechanisms of oseltamivir and zanamivir. The binding of oseltamivir to the active site of neuraminidase requires a conformational change, whereas this change is not required for zanamivir (395). It is interesting to note that the *in vitro* triple combination of oseltamivir, amantadine, and ribavirin is highly synergistic against the A/2009/H1N1 virus (411).

Oseltamivir resistance has been reported in patients without prior exposure to oseltamivir (Table 15), and it can develop quickly after oseltamivir treatment (272). It is especially common in immunosuppressed hosts, in whom a higher viral load is expected due to the poor control by the host immune response. The nosocomial spread of an oseltamivir-resistant influenza virus has been documented in a clinical setting (392). Other mutations (E119G, E119V, and I222V) have been found to confer oseltamivir resistance *in vitro*, but they have not been reported in clinical strains (445).

The S247N mutation, which was found in 30% of the specimens from northern Australia, results in 6-fold and 3-fold reductions in susceptibility to oseltamivir and zanamivir, respectively (266). Despite this low level of resistance *in vitro*, there were reports of clinical treatment failure (410, 521). The combined S247N and H275Y mutations resulted in a >5,000-fold reduction in oseltamivir susceptibility. A structural analysis using computer modeling suggested that the S247N mutation pushes the E277 residue deeper into the drug binding pocket, resulting in reduced oselta-

**TABLE 13** Clinical or virological response of different patient groups started on antiviral therapy at different time after onset of A/2009/H1N1 infection<sup>a</sup>

| Country(ies) | Population                                     | Day of illness at initiation of antiviral treatment                 | Findings                                                                                                                  | Reference |
|--------------|------------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-----------|
| China        | Hospitalized patients during containment phase | ≤2 vs >2 (no. of patients in each group not mentioned)              | Late treatment associated with longer viral shedding duration (OR, 4.46)                                                  | 58        |
| Singapore    | Hospitalized patients during containment phase | ≤2 ( <i>n</i> = 36) vs >2 ( <i>n</i> = 34)                          | Patients with earlier treatment had shorter viral shedding duration                                                       | 357       |
| Hong Kong    | Hospitalized patients during containment phase | ≤2 ( <i>n</i> = 83) vs >2 ( <i>n</i> = 35)                          | Patients with earlier treatment had faster viral load reduction and shorter viral shedding duration                       | 350       |
| USA          | Hospitalized patients                          | ≤2 ( <i>n</i> = 75) vs >2 ( <i>n</i> = 120)                         | Late treatment was associated with ICU admission or death                                                                 | 281       |
| USA          | Hospitalized patients                          | ≤2 ( <i>n</i> = 36) vs >2 ( <i>n</i> = 40)                          | Early treatment was associated with shorter lengths of stay in hospital ( <i>P</i> = 0.03)                                | 75        |
| Spain        | Hospitalized patients                          | ≤3 ( <i>n</i> = 297) vs >3 ( <i>n</i> = 288)                        | Early treatment was associated with nonsevere disease (OR, 0.32)                                                          | 529       |
| Canada, USA  | Solid-organ transplant recipients              | ≤2 ( <i>n</i> = 90) vs >2 ( <i>n</i> = 125)                         | Late treatment was associated with ICU admission ( <i>P</i> = 0.007)                                                      | 322       |
| Argentina    | Solid-organ transplant recipients              | NA                                                                  | Late treatment was associated with more severe disease ( <i>P</i> = 0.008)                                                | 482       |
| Vietnam      | Hospitalized patients                          | NA                                                                  | Estimated median clearance times between 2.6 and 2.8 days posttreatment for illness-to-treatment intervals of 1 to 4 days | 257       |
| USA          | Pregnant women                                 | ≤2 ( <i>n</i> = 30) vs >2 ( <i>n</i> = 30)                          | Late treatment was associated with ICU admission or death (RR, 4.3)                                                       | 363       |
| USA          | Pregnant women                                 | ≤2 ( <i>n</i> = 30) vs 3-4 ( <i>n</i> = 14) vs ≥5 ( <i>n</i> = 9)   | Late treatment was associated with severe illness ( <i>P</i> = 0.002)                                                     | 128       |
| France       | Pregnant women                                 | <3 ( <i>n</i> = 237) vs 3-5 ( <i>n</i> = 39) vs >5 ( <i>n</i> = 23) | Late treatment was associated with severe disease (3-5 days, adjusted OR = 4.78; >5 days, adjusted OR = 61.24)            | 159       |
| USA          | Children                                       | ≤2 ( <i>n</i> = 5) vs >2 ( <i>n</i> = 6)                            | No difference in length of ICU stay                                                                                       | 361       |
| China        | All ages                                       | <5 ( <i>n</i> = 2,858) vs ≥5 ( <i>n</i> = 3,807)                    | Increased risk of severe disease (OR, 1.42)                                                                               | 574       |

<sup>a</sup> ICU, intensive care unit; OR, odds ratio; RR, relative risk; NA, not applicable.

mir binding. Additionally, the I223R and I223K substitutions in neuraminidase can result in reduced susceptibility to zanamivir (410, 521).

There has been controversy regarding the fitness and virulence of resistant mutants. The H275Y mutation in the A/2009/H1N1 virus has been shown to reduce viral replication *in vitro* (46). However, H275Y mutants were as virulent as wild-type viruses in mouse and ferret models, with no reduction in transmissibility (230, 383).

### Safety of Antivirals

Neuraminidase inhibitors are relatively safe. Reports from Japan suggested that oseltamivir may be associated with neurotoxicity in adolescents. Oseltamivir and its metabolites may have excitatory effects on the central nervous systems of rats, which may account for some of the neuropsychiatric side effects that have been observed in the past few years with more widespread use of the drug (278). The safety of these agents in infants and pregnant women is unknown. Oseltamivir and zanamivir are currently approved for children over 1 and 7 years of age, respectively. Limited studies have demonstrated that oseltamivir is safe in infants under 1 year of age (136, 310). Currently, there is no evidence that oseltamivir is teratogenic, but it is classified as FDA cate-

gory C due to the lack of data. Additional safety data should be collected due to the increasing use of intravenous zanamivir and peramivir (54, 314, 495).

### Passive Immunotherapy

There is no definitive evidence that antivirals work in severe cases when patients present late after the onset of symptoms. Thus, modulation of the host immune response by dampening down the proinflammatory damage with steroids, statins, and macrolides has been studied (Table 16); however, none of these drugs were tested in the setting of a randomized controlled trial. Convalescent-phase plasma is another potentially useful form of therapy for severe influenza due to the influenza A virus. Convalescent-phase plasma has been used in one patient with A/H5N1 infection, with good clinical and virological responses (585). A meta-analysis of the use of convalescent-phase blood products during the Spanish influenza pandemic showed that there was a survival benefit in patients who received this treatment, and the benefit was greater if treatment was initiated within 4 days of the onset of pneumonia (367). Based on these findings, a prospective cohort study was conducted on the treatment of patients with severe A/2009/H1N1 infection using convalescent-phase plasma collected from sur-



**TABLE 14** Use of intravenous antivirals for A/2009/H1N1 infection<sup>a</sup>

| Country | Population                                                                                                                                                | Findings                                                                                                                                                                                                   | Reference |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| USA     | Case series; 20 adults and 11 children had rapidly progressing radiologically confirmed viral pneumonia with respiratory failure                          | Peramivir was administered for 1–14 days (median duration, 10 days); survival rate at 14 days, 76.7%; survival rate at 28 days, 66.7%; survival rate at 56 days, 59%; no reports of serious adverse events | 253       |
| USA     | 33 received peramivir; 8 received i.v. zanamivir                                                                                                          | 27.7% mortality                                                                                                                                                                                            | 186       |
| Germany | 2 patients with ARDS and ICU admission (M/39 yr, with no underlying disease, and M/49 yr, a smoker with diabetes mellitus, nephropathy, and hypertension) | Both patients improved after i.v. zanamivir                                                                                                                                                                | 238       |
| UK      | F/22 yr with neutropenic fever after chemotherapy for Hodgkin's disease                                                                                   | High level of H1N1 RNA still detected after 6 days of oseltamivir and nebulized zanamivir; after change to i.v. zanamivir and methylprednisolone, patient improved                                         | 303       |
| Germany | M/2 yr, after liver transplant for Caroli's disease                                                                                                       | During i.v. zanamivir treatment, no clinical improvement but liver and renal function worsened                                                                                                             | 151       |
| USA     | F/18 mo, undergoing allogeneic matched related stem cell transplant                                                                                       | Regimen was well tolerated and associated with a decrease in viral burden                                                                                                                                  | 160       |
| Canada  | Case report; M/50s, allogeneic stem cell transplant recipient                                                                                             | Proportion of H275Y mutant increased during peramivir treatment but decreased after treatment stopped                                                                                                      | 456       |
| Japan   | Case report; F/40s, diabetic nephropathy                                                                                                                  | Rapid improvement in respiratory failure after peramivir                                                                                                                                                   | 405       |
| USA     | Case report; F/40s, multiple myeloma                                                                                                                      | Respiratory failure despite oseltamivir, amantadine, ribavirin, and IVIG; improvement after peramivir                                                                                                      | 57        |
| USA     | 2 critically ill adult patients on CVVHDF; daily infusion of 600 mg over 30 min while on CVVHDF; study to describe pharmacokinetics                       | $C_{\max}$ = 18,400 and 20,300 ng/ml; plasma $t_{1/2}$ = 7.6 and 3.7 h; $V_d$ = 0.51 and 0.54 liters/kg                                                                                                    | 24        |

<sup>a</sup> ARDS, acute respiratory distress syndrome;  $C_{\max}$ , maximum concentration; CVVHDF, continuous venovenous hemodiafiltration; F, female; ICU, intensive care unit; i.v., intravenous; IVIG, intravenous immunoglobulin; M, male;  $t_{1/2}$ , half-life;  $V_d$ , volume of distribution.

viving patients (264, 557). In this study, a number of practical limitations were encountered during the collection of convalescent-phase plasma by apheresis (548). These limitations included the failure of donors to meet blood donation eligibility criteria, failed laboratory tests, insufficient neutralizing antibody titers, and the inability of donors to attend the apheresis appointment. Despite these limitations, 276 liters of convalescent-phase plasma were collected from more than 60% of the potential donors with sufficient neutralizing antibody titers. None of the patients in the treatment group developed adverse effects after convalescent serotherapy. The mortality in the treatment group was significantly lower than that in the non-treatment group. In addition, the reductions of viral load and the corresponding cytokine/chemokine were greater in the treatment group. These two studies demonstrated that passive immunotherapy was feasible (344) and that the treatment of severe A/2009/H1N1 infection with convalescent-phase plasma appeared safe and effective, which has important implications for treatment strategies using passive immunotherapy in future pandemics. However, these findings should be confirmed in randomized controlled trials.

Broadly neutralizing antibodies against the stalk of H and the M2 ectodomain may be important in passive immunization. Humanized neutralizing anti-M2e monoclonal antibodies were shown to protect against lethal challenge by H5N1 and H1N1 viruses (210). A monoclonal antibody that targeted the conserved F subdomain of HA2 has been shown to protect mice and ferrets from both H1N1 and H3N2 infections (122). A randomized controlled trial of treatment with specific human monoclonal antibodies or hyperimmune intravenous immunoglobulin in patients with severe influenza infection is warranted.

### Controversial Use of Steroids and Other Immunomodulators

The use of corticosteroids in the treatment of A/2009/H1N1 influenza is controversial because no randomized controlled trials have been conducted. Some studies have shown a beneficial effect of corticosteroids, whereas other studies showed no effect or even a detrimental effect (Table 16). One study demonstrated that the risk of death was higher in the corticosteroid group only in patients who received early administration of corticosteroids within 3 days of mechanical ventilation (48). Case reports have indicated that therapy with etoposide and betamethasone was beneficial in a patient with hemophagocytic lymphohistiocytosis (251). High-dose *N*-acetylcysteine has been shown to reduce the inflammatory response (327). Statins that were started before or after influenza infection were not associated with better outcomes (43, 528). Animal and cell culture models have suggested that agents such as paracetamol and mesalazine may reduce damage, but these findings remain to be confirmed in humans (334, 549, 581). In mice, activation of the innate immune system by PIKA, a chemical analog of dsRNA, is associated with a reduction in viral replication in the respiratory tract (333). Recently, an anti-inflammatory immunomodulator, a sphingosine analog, was successfully used in combination with oseltamivir to improve the survival of A/2009/H1N1-infected mice by significantly blunting the cytokine storm and tissue injury. In contrast to steroids, this novel approach does not suppress the protective adaptive immune response of the host against the invading virus and is a more logical approach for treating patients with severe disease (532).

TABLE 15 Oseltamivir resistance of A/2009/H1N1 virus<sup>a</sup>

| Country (total no. of isolates tested)                    | Population with resistance gene                                                                                                                                          | Mutation(s) (N1 numbering), IC <sub>50</sub> (nM) (if available) | Change of antiviral; outcome                        | Reference |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|-----------------------------------------------------|-----------|
| Worldwide (close to 1,000)                                | 3 patients (Denmark, Japan, Hong Kong) <sup>b</sup>                                                                                                                      | NM                                                               | No change; survived                                 | 552       |
| Worldwide (NA)                                            | 39 patients (32 with information); 16 associated with treatment, 7 immunosuppressed, 13 with chemoprophylaxis, 3 with no history of oseltamivir treatment or prophylaxis | H275Y                                                            | NM                                                  | 551       |
| USA (3,359)                                               | 0.7% were resistant                                                                                                                                                      | H275Y                                                            | NA                                                  | 219       |
| USA (6,740)                                               | 0.5% were resistant; among the patients with resistant strains, 76% were immunocompromised and 86% received oseltamivir before specimen collection                       | H275Y                                                            |                                                     | 209       |
| Canada (804)                                              | 5 patients (0.6%)                                                                                                                                                        | H275Y                                                            | NA                                                  | 362       |
| Australia (71 pretreatment and 25 posttreatment isolates) | 3 in posttreatment samples (M/62 yr allogeneic stem cell transplant recipient, M/47 yr immunocompetent, F/19 mo with malignancy) <sup>c</sup>                            | H275Y                                                            | M/47 yr received alternative therapy; all recovered | 533       |
| Australia (17 with HSCT, 15 with malignancy)              | 4 patients <sup>c</sup>                                                                                                                                                  | H275Y                                                            | Changed to zanamivir in 1 patient; 2 died           | 513       |
| Scotland (1,640)                                          | 15 immunocompromised patients <sup>c</sup>                                                                                                                               | H275Y                                                            | NM                                                  | 239       |
| UK (2,864)                                                | 27 cases                                                                                                                                                                 | H275Y                                                            | NA                                                  | 325       |
| Italy (186)                                               | F/2 yr with acute lymphoid leukemia <sup>c</sup>                                                                                                                         | H275Y                                                            | No change; survived                                 | 56        |
| Italy (31)                                                | 3 strains                                                                                                                                                                | H275Y                                                            | NA                                                  | 450       |
| Germany (1,570)                                           | 8 patients                                                                                                                                                               | H275Y, 200-400                                                   | NA                                                  | 163       |
| Spain (1,229)                                             | 8 patients (1.93%)                                                                                                                                                       | H275Y                                                            | NA                                                  | 338       |
| Spain (14)                                                | 2 patients (M/40 yr with HIV, M/49 yr with hematological malignancy)                                                                                                     | H275Y                                                            | M/49 yr; oseltamivir was changed to zanamivir       | 7         |
| Asia-Pacific region (1,488)                               | Singapore, 3.1%; Australia, 1.3%                                                                                                                                         | H275Y                                                            | NA                                                  | 267       |
| Japan (75)                                                | 1 strain                                                                                                                                                                 | H275Y, 46                                                        | NA                                                  | 300       |
| Japan (253)                                               | 3 patients                                                                                                                                                               | H275Y (2 patients), N295S (1 patient)                            | NA                                                  | 394       |
| United Arab Emirates (96)                                 | 8-yr-old child who received prophylactic oseltamivir                                                                                                                     | H275Y                                                            | NM                                                  | 5         |
| Hong Kong (95)                                            | F/16 yr from San Francisco, CA, hospitalized for isolation <sup>d</sup>                                                                                                  | H275Y, <sup>e</sup> 197.5                                        | No change; survived                                 | 93        |
| Mexico (692)                                              | 8-mo-old patient                                                                                                                                                         | H275Y, 27.3                                                      | NM                                                  | 454       |
| USA                                                       | 2 adolescent girls attending summer camps who received oseltamivir prophylaxis <sup>c</sup>                                                                              | H275Y, I223V                                                     | Changed to zanamivir in 1 patient; survived         | 71        |
| Canada                                                    | M/59 yr with COPD and taking prednisolone 5mg daily (emergence during prophylaxis) <sup>c</sup>                                                                          | H275Y, >400                                                      | NM                                                  | 23        |
| USA (4)                                                   | Nosocomial transmission of oseltamivir-resistant A/2009/H1N1 in immunocompromised patients                                                                               | H275Y                                                            | NA                                                  | 98        |
| Singapore                                                 | F/28 yr, developed resistance within 48 h of oseltamivir treatment                                                                                                       | H275Y                                                            | NA                                                  | 272       |

<sup>a</sup> Case reports of oseltamivir-resistant influenza infection not shown in this table are in references 11, 88, 160, 168, 194, 197, 257, 259, 384, 488, 505, and 590). COPD, chronic obstructive pulmonary disease; F, female; HSCT, hematopoietic stem cell transplant; M, male; NA, not applicable; NM, not mentioned; IC<sub>50</sub>, 50% inhibitory concentration.

<sup>b</sup> History of oseltamivir therapy was unknown.

<sup>c</sup> Oseltamivir resistance developed after oseltamivir therapy.

<sup>d</sup> Oseltamivir resistance occurred before oseltamivir therapy.

<sup>e</sup> Quasispecies.

## EXTRACORPOREAL MEMBRANE OXYGENATION

Of the hospitalized patients with A/2009/H1N1 infection, 10% to 44% required intensive care, and a significant proportion of these patients developed ARDS. Moreover, those patients with A/2009/

H1N1 infection who required intensive care were significantly younger than and had fewer underlying medical conditions than patients with seasonal influenza infection. Because of the encouraging results from the CESAR (conventional ventilatory support

**TABLE 16** Use of immunomodulators for treatment of severe A/2009/H1N1 infection<sup>a</sup>

| Country(ies)                | Population                                                                  | Immunomodulator                                                                                                                                                                                                                                                                                               | Concomitant therapy                                                                           | Findings                                                                                                                                                                                                                                                              | Reference |
|-----------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| South Korea                 | 245 ICU patients; 107 received corticosteroid                               | Corticosteroid (given within 2 days of ICU admission): hydrocortisone, 50%; methylprednisolone, 38%; other corticosteroid, 12%                                                                                                                                                                                | No significant difference in oseltamivir use between steroid group and nonsteroid group       | 90-day mortality significantly higher in those with corticosteroid (58%) than those without (27%); corticosteroid group more likely to have superinfection                                                                                                            | 306       |
| France                      | 208 ICU patients with ARDS; 83 received corticosteroid                      | Corticosteroid (60.2% given within 3 days of ICU admission): hydrocortisone, 57.8%; methylprednisolone, 37.3%; prednisolone, 4.8%                                                                                                                                                                             | No significant difference in oseltamivir use between steroid group and nonsteroid group       | Corticosteroid associated with death (adjusted hazard ratio, 2.82); subgroup analysis showed that only early administration of corticosteroids within 3 days of mechanical ventilation was associated with the increased risk of death, not administration at >3 days | 48        |
| Spain, Brazil, UK, Portugal | 220 ICU patients; 126 received corticosteroid                               | Corticosteroid at ICU admission: >24 mg/day methylprednisolone or >30 mg/day prednisolone                                                                                                                                                                                                                     | No significant difference in oseltamivir use between steroid group and nonsteroid group       | Corticosteroid treatment associated with increased risk of hospital acquired pneumonia (OR = 2.2) and ICU mortality (OR = 3.8)                                                                                                                                        | 377       |
| China                       | 155 hospitalized patients; 52 received corticosteroid                       | Corticosteroid: median methylprednisolone dose, 80 mg (or equivalent dose)                                                                                                                                                                                                                                    | 80.6% of all patients received oseltamivir                                                    | Corticosteroid treatment associated with a trend toward higher hospital mortality ( $P = 0.052$ )                                                                                                                                                                     | 560       |
| Argentina                   | 13 adult patients with ICU admission; all received corticosteroid           | Corticosteroid (intravenous hydrocortisone 100 mg every 8 h until ICU discharge, then 100 mg every 12 h for 7 days and 100 mg daily for 7 days); for patients with ARDS, hydrocortisone was changed to methylprednisolone at 1 mg/kg/day for 14 days, then 0.5 mg/kg/day for 7 days, then tapered over 6 days | High-dose oseltamivir (150 mg twice daily for 5 days, then 75 mg twice daily for 3 to 5 days) | Significant reduction in CRP, APACHEII, lung injury, and SOFA; 15% mortality                                                                                                                                                                                          | 453       |
| Hong Kong                   | 93 patients (20 patients with plasma therapy)                               | Convalescent-phase plasma                                                                                                                                                                                                                                                                                     | All received high-dose oseltamivir                                                            | Multivariate analysis showed that plasma treatment reduced mortality (OR = 0.2)                                                                                                                                                                                       | 264       |
| Australia                   | 5 patients                                                                  | IVIG                                                                                                                                                                                                                                                                                                          | 3 patients received ECMO, 2 patients received intravenous zanamivir                           | 3 patients had clinical improvement; 2 patients had respiratory deterioration following IVIG                                                                                                                                                                          | 207       |
| UK                          | 1,520 patients                                                              | Statin                                                                                                                                                                                                                                                                                                        | NM                                                                                            | No significant association between preadmission statin use and severity of outcome                                                                                                                                                                                    | 43        |
| Spain                       | 197 patients with pneumonia; 68 patients received anti-inflammatory therapy | Corticosteroid (37), macrolide (31), statin (12)                                                                                                                                                                                                                                                              | NM                                                                                            | None of the therapies were associated with lower risk of developing severe disease                                                                                                                                                                                    | 528       |

<sup>a</sup> APACHEII, Acute Physiology and Chronic Health Evaluation II score; ARDS, acute respiratory distress syndrome; CRP, C-reactive protein; ECMO, extracorporeal membrane oxygenation; ICU, intensive care unit; IVIG, intravenous immunoglobulin; NM, not mentioned; OR, odds ratio; SOFA, sequential organ failure assessment score.

versus extracorporeal membrane oxygenation [ECMO] for severe adult respiratory failure) trial, ICUs from different centers around the world (86, 139, 185, 358, 414) have treated patients using ECMO, with promising results. The rate of ECMO use for all mechanically ventilated patients with A/2009/H1N1 pneumonia ranged from 4% to 9%. The survival rates for patients who underwent ECMO and mechanical ventilatory support were similar across the studies and ranged from 66% to 86%, with no severe complications associated with the use of ECMO. In addition, a study in Italy demonstrated the feasibility of transferring critically ill A/2009/H1N1-infected ARDS patients to ECMO centers for treatment, with a high survival rate (433). The roles of different adjunctive therapies that combine immunotherapy and ECMO support for the treatment of patients with severe A/2009/H1N1 infection should be assessed in future studies.

## VACCINES

Despite the best available treatments, the mortality from severe A/2009/H1N1 infection is substantial. Prevention by vaccination is crucial. In addition to the prevention of pulmonary complications, influenza vaccination can prevent medical catastrophes triggered by influenza, such as myocardial infarction and stroke, especially when combined with pneumococcal vaccination (263).

The greatest setback in the 2009 influenza pandemic was the non-availability of the vaccine until late 2009 in most countries. Thus, the summer peak in tropical areas and the winter peak in the Southern Hemisphere during August 2009 passed before vaccines were administered. Innovations, such as cell-based whole-virion inactivated vaccines and dose-sparing adjuvants (17, 129, 519), boosted the vaccine supply but were unable to shorten the interval between the first detection of the virus and the availability of the vaccine in the market (249, 479). A hemagglutination inhibition titer of 1:40 has been achieved in more than 90% of vaccinees, although children less than 11 years of age or adults greater than 61 years of age had poorer responses (588). Other factors that are associated with a poor response to vaccination include immunosuppressed states or chronic diseases. Adjuvants and two-dose regimens may improve immunogenicity. Dose sparing by the use of the intradermal route or new adjuvants was shown to have equivalent efficacy to that of a normal dose of vaccine (508). Vaccine effectiveness, as defined by the reduction in the risk of laboratory-confirmed infection between vaccinated and nonvaccinated individuals, has been estimated to be approximately 70% (235). It should be noted that live attenuated vaccines induce a lower antibody response in adults than inactivated vaccines (62),

and vaccine recipients have had lower antibody titers than individuals with natural infection (84).

Prior seasonal influenza vaccination has been found to worsen the outcomes of patients with A/2009/H1N1 infection (283, 478). The increased risk may be due to the lack of cross-reactive immunity induced by natural seasonal influenza infection (385). However, other clinical studies have demonstrated contradictory findings, with no apparent increase in the severity of pandemic A/2009/H1N1 infection in individuals who received a prior seasonal influenza vaccine (126, 192, 289). It has been shown that prior seasonal influenza vaccination can increase the levels of antibodies against neuraminidase in the elderly (374). Furthermore, cross-reactive T cells induced by the seasonal influenza vaccine may confer some degree of protection against pandemic A/2009/H1N1 infection (258). The acceptance of the pandemic influenza vaccine is low, partly due to the concerns regarding safety, such as the development of Guillain-Barre syndrome after vaccination (330, 466, 467, 512). Nevertheless, both premarketing trials and postmarketing surveys showed that the risk of developing Guillain-Barre syndrome in individuals who receive the pandemic influenza vaccine is similar to that in the general population (76, 353).

### INFECTION CONTROL IN THE COMMUNITY AND HOSPITALS

Before the emergence of the A/2009/H1N1 pandemic, nonpharmacological interventions with social distancing, such as school closures, had been evaluated in modeling (26, 175) and epidemiological (38, 125, 256, 375) studies. School closures were practiced in North America and Australia during the 2009 pandemic (40, 121, 165) because of a high clinical attack rate that ranged from 10% to 60% (133, 221, 248, 275, 347, 514) (Table 3). School closures were associated with a 65% reduction in the mean total number of contacts for each student as reported in a retrospective questionnaire survey in the United Kingdom (279). However, a late school closure in which 27% of the students already had symptoms had no significant impact on the spread of infection (63). Therefore, school closures should begin once the threshold for daily case counts is exceeded (228). In Hong Kong, kindergartens and primary schools were closed when local transmission of the A/2009/H1N1 virus was identified, followed shortly afterwards by secondary school closures for summer vacations. The transmission of the A/2009/H1N1 virus was estimated to be reduced by 25% (556).

Household transmission of the A/2009/H1N1 virus was reported to have an attack rate of up to 45% (Table 3) (59, 64, 124, 182, 284, 318, 393, 428, 436, 550, 558, 571). The transmissibility of the A/2009/H1N1 virus in the household setting was broadly similar to that of seasonal influenza A viruses (124). The attack rate was highest among children and adolescents. Household contacts less than 18 years of age were approximately 15 times more likely to be infected than older contacts (348). When oseltamivir was used as postexposure prophylaxis, the household secondary attack rate was reduced from 26.1% to 0.6% (318). However, whether the early treatment of the index person with oseltamivir can reduce secondary infections in the household setting remains to be determined (203). Hand hygiene and face masks appeared to prevent household transmission of influenza virus when implemented within 36 h of symptom onset in the index patient (123). The wearing of face masks was well tolerated by children less than 14 years of age (494). Frequent hand washing has been shown to

reduce surface contamination by influenza virus in the household (475). Simple nonpharmaceutical interventions may mitigate pandemic influenza if compliance can be ensured.

The nosocomial transmission of the A/2009/H1N1 virus was reported to have an attack rate of 10% to 45% in health care settings (531), including acute-care hospitals and long-term care facilities (10, 74, 114, 172). Among the 30 patients who acquired A/2009/H1N1 infection during hospitalization in the United Kingdom, 8 (27%) of the patients died. Most of them had an underlying malignancy or immunosuppressive conditions (166). For health care workers (HCWs), asymptomatic infection as evidenced by seroconversion was observed in 28 (9.6%) of 290 nurses, especially in nurses who worked in the isolation wards (100). Therefore, an infection control program that consisted of multiple coherent measures was proposed to minimize the nosocomial transmission of the A/2009/H1N1 virus during the early phase of the pandemic in Hong Kong. This program included several open staff forums that achieved high attendance, the early recognition of index cases among inpatients by liberal testing, the early relief of sick staff from work, directly observed hand hygiene practices during outbreaks, and the monitoring of compliance with infection control practices. With these measures, only 4 (0.48%) of 836 persons who were exposed to laboratory-confirmed patients and staff with A/2009/H1N1 infection were virologically confirmed to have A/2009/H1N1 infection. Not wearing a surgical mask, either by exposed persons during contact with index cases (4/4 versus 264/832;  $P = 0.010$ ) or vice versa (4/4 versus 300/832;  $P = 0.017$ ), was found to be a significant risk factor for the nosocomial acquisition of A/2009/H1N1 infection (108). Additionally, this finding was observed in a study in Singapore that demonstrated that the incidence of pandemic A/2009/H1N1 infection remained low in HCWs who wore surgical masks (10). In contrast, surgical mask use was not consistently practiced in the United States, where the Centers for Disease Control and Prevention received reports of 70 HCWs with A/2009/H1N1 infection from 22 states. Of these cases, 35 (50%) were classified as being infected in a health care setting. Of the 23 HCWs infected by patients, only 20% reported using an N95 or surgical mask during all patient care practices (546).

Good adherence to infection control practices may reduce potential occupational exposure to the A/2009/H1N1 virus in the hospital setting. In a seroprevalence study of the A/2009/H1N1 antibodies in 599 HCWs after the first wave in Hong Kong, only 12% of the HCWs who did not receive the pandemic vaccine had antibody titers of  $\geq 1:40$  as detected by a viral neutralization assay (587). Influenza vaccination has been strongly advocated to protect both patients and HCWs, especially HCWs who work in ICUs with high-risk aerosol-generating procedures, such as intubation, resuscitation, and the delivery of aerosolized medications (138). However, the A/2009/H1N1 vaccination uptake was less than 50% in a group of HCWs who worked in the critical care and theater settings in the United Kingdom (430), whereas the A/2009/H1N1 vaccination rate of HCWs was just over 30% in France (503) but less than 20% in Spain and Italy (9, 145). The A/2009/H1N1 vaccination uptake rate is higher in the United States. Among the HCWs who worked at a facility where vaccination was required by their employer, 98.1% were vaccinated, whereas only up to 70% of HCWs were vaccinated without this employer requirement (70). In addition, a high level of acceptance with a voluntary uptake rate of up to 70% was observed in Canada (292). However, the A/2009/



H1N1 vaccine appeared to have suboptimal immunogenicity in several HCWs. Of 104 HCWs who received the A/2009/H1N1 vaccine in Hong Kong, only 42% had an antibody titer of  $\geq 1:40$  as detected by viral neutralization assay. The proportion of HCWs with an antibody titer of  $\geq 1:40$  significantly decreased with age (586). Additional studies are required to confirm whether the A/2009/H1N1 vaccine maintains high efficacy and effectiveness in HCWs and prevents the nosocomial transmission of the A/2009/H1N1 virus.

## CONCLUSION

The lessons of the human pandemic of A/2009/H1N1 virus in 2009, the epidemic of SARS coronavirus in 2003, and the poultry epidemic of influenza A H5N1 virus since 1997 suggest that preparedness against agents of emerging infectious diseases that jump from animals to humans must continue. Rapid economic growth in many developing areas of the world has led to an increasing demand for animal proteins, such as pork. Large numbers of swine and other food animals are reared with antibiotic-loaded feeds in overcrowded conditions. Inadequate biosecurity measures allow for the continued jumping of novel influenza virus reassortants or multidrug-resistant bacteria from swine to humans. The potential combination of influenza and antibiotic-resistant bacterial infections could prove disastrous should future pandemics occur. Improved human and animal surveillance programs are essential for a rapid response against this occurrence. Currently, no bioinformatic or *in vitro* tests can reliably predict the virulence or clinical severity of potential new pandemic viruses. Transgenic humanized animal models should be investigated to address this important issue for better resource allocation.

The superiority of the nucleic acid amplification test has been demonstrated in the clinical management of A/2009/H1N1 infection. A multiplex system coupled with microfluidic technology that uses a small amount of clinical samples can enhance the robustness and application of nucleic acid amplification in field settings. More antiviral agents that attack the different parts of the viral life cycle should be developed to overcome antiviral resistance. Novel immunomodulatory agents that dampen detrimental inflammatory responses without hindering the protective immune response of the host should be developed and tried in those late presenters with severe diseases due to overly activated cytokines and chemokines. In future pandemics, instead of waiting for a new vaccine that is based on the pandemic virus, heterologous protection should be achieved with a readily available universal vaccine that is based on the highly conserved stalk region of H and the ectodomain of M2. This universal vaccine will overcome the delay between the detection of the new virus and the availability of the vaccine in the market from mass production. Strategies that accelerate a vaccine-induced immune response should be developed. The influenza virus is highly unpredictable, and we should always be open-minded when facing the unknown.

## ACKNOWLEDGMENTS

We acknowledge research funding from the Providence Foundation, Ltd., in memory of the late Lui Hac Minh, the Shaw Foundation for Infectious Diseases Research, the HKSAR Research Fund for the Control of Infectious Diseases of the Health, Welfare and Food Bureau, and the Consultancy Service for Enhancing Laboratory Surveillance of Emerging Infectious Disease for the HKSAR Department of Health.

We thank Chung-Ying Leung, Consultant Pathologist, United Christian Hospital, for providing the histopathology pictures from patients who died of A/2009/H1N1 influenza.

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